

Unpacking the Invisible Barriers: A Mixed-Methods Usability Evaluation of Learning Management Systems in Bangladeshi Universities

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
June 2026.

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Declaration

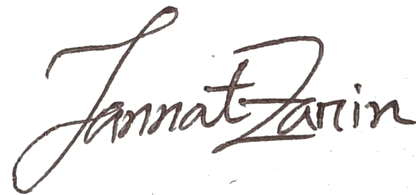
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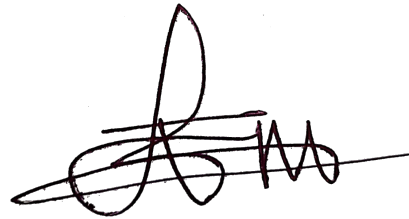
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Abstract

Learning Management Systems (LMS) have become more popular among the Bangladeshi universities during this digitalization era. Still, institutional barriers continue to prevent the successful utilization of these platforms by many students. A exploratory sequential mixed-methods approach was adopted as a way to investigate such issues in detail. We surveyed selected student groups quantitatively in the form of structured surveys, through which we collected the finer data on the perceptions of usability and problematic features in specific cases as well as the system usability in general using the System Usability Scale (SUS). Among these features, there are such aspects as the complexity of navigation, the problems with notifications, submission procedures, platform fragmentation, technical and infrastructural constraints, training deficiencies, instructor proficiency, and role perception. These questionnaires contained open-ended questions in which the participants were asked to air out the usability-related issues that are subtle using their own words. A thematic analysis of thirty-four interviews with undergraduate students, graduate students and faculty members produced seven interconnected usability themes, and a complementary survey of 222 active LMS users was used to test seven corresponding hypotheses through chi-square and one-sample t-tests. All seven hypotheses were supported, confirming that these obstacles are systemic and statistically significant rather than incidental. In this way, obstacles in using LMS in the Bangladeshi higher education environment were identified and addressed. These empirical findings and results were used to develop a functional, high-fidelity prototype and to give practical, evidence-based redesign recommendations that are applicable to system developers and administrators and consider the unique pedagogical, sociocultural, and infrastructural dimensions of the Bangladesh educational context. This paper intends to transforms digital learning environments into actually usable, accessible and pedagogically useful instruments by systematically locating and analyzing the often opaque obstacles that are integrated into LMS interfaces and processes.

Keywords: Learning Management Systems (LMS), Higher education institutions, Digital learning platforms, Bangladesh, Usability, System Usability Scale (SUS), Digital literacy, Pedagogical design, Sociocultural factors, Optimization; Mixed-methods, Thematic analysis, Platform fragmentation, UI, Qualitative, Quantitative.

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Chapter 1

Introduction

1.1 Background

Digital education has revolutionized how institutions of higher learning provide and administer learning, and the e-learning market in Asia is increasing at a rate of about 17.3 percent per year [62]. This increase reflects a larger generational change towards the use of technology and a slow shift in the teaching and learning process out of the traditional classroom setting to digitally mediated learning spaces [57]. This shift has become a critical one with the onset of the COVID-19 pandemic and has made it a necessity for institutions around the world to move their teaching online in a matter of days [62]. In this landscape, the use of Learning Management Systems (LMS) is now the norm rather than the exception, with systems such as Moodle, for example, accounting for an estimated 291 million users across 241 countries, providing a single central point for content delivery and communication, both synchronously and asynchronously [23]. As a result, LMS platforms have become a core part of modern-day higher education.

Although these platforms have been heavily institutionalized, there is a long-standing paradox in the use of these platforms in higher education institutions (HEIs): the systems are extensively used but not fully exploited [62]. Resistance by users has been cited as the biggest obstacle to successful implementation again and again, and in most instances the LMS is being used as a digital library of course materials and not as a learning environment [62]. This is supported by empirical research; Simon et al. found that the majority of students and instructors viewed Moodle as a file-sharing platform, and faculty in general found it difficult to use because of a formal lack of training [62]. Koca and Özcan also presented a moderate score of usability for faculty ($SUS = 63.85$), and uncovered two main problems that faculty experienced, which were poor support and insufficient training [57]. This disconnect between adoption and meaningful use is an indicator that the root cause of the issue is not the availability of the technology but rather the quality of the user experience that the technology offers.

The usability and design features that would stop meaningful interaction and decrease the likelihood of user retention are the reasons for lack of meaningful use of LMS platforms [57]. The interface issues that are always found to be the main causes of user frustration and dissatisfaction are complex menus, being lost in nav-

igation, and overloaded and cluttered screens [23], [57]. Surveys using standardized tools have yielded conflicting results on the following questions: Berényi et al. documented a fairly high usability rating of Moodle among Hungarian learners (SUS = 85.04) but reported that notification and collaboration tools were mostly ignored [52], and Ibrahim and Aziz rated a university platform below average or poor in most user-experience metrics [21]. Alkoblan and Al-Wadud showed that the objective and subjective testing methods revealed 31 usability defects, almost three times as many as traditional single-method testing, especially in system visibility and user control [25]. In line with these results, Al-Fraihat et al. explained that negative student experience was caused by an unattractive design and low-quality content [54], and Faghih et al. claimed that user interface design was not a cosmetic issue but a central factor of engagement and learning outcomes [1].

Another weakness in the discipline is the way that usability is assessed. Most of the research that has been done is based on general human-computer interaction tools, most prominently the System Usability Scale, which is not only a measure of technical usability but also ignores the pedagogical and sociocultural aspects of educational technology [59]. To fill this gap, Lu, Schmidt, and Shin created and tested a multidimensional assessment instrument that incorporates technological, pedagogical, and sociocultural usability, and critiques the technical tunnel vision of SUS-based practice explicitly [59]. Systematic reviews resonate with this point; Secaira et al. found that ease of use and user satisfaction are prevalent in the LMS evaluation literature and demanded multidimensional, context-specific quality models [61]. Almusharraf also demonstrated that perceptions of users change with experience and that cultural and contextual influences determine satisfaction and acceptance [28], and that any plausible LMS assessment should consider the context within which the system is applied, rather than treat usability as a technical attribute of the system.

In fact, in resource-constrained settings such as Bangladesh, where present internet access is not very secure, device availability is limited, and the level of digital literacy varies widely, it is a legitimate requirement to have a simplified, robust system, as minor usability problems are compounded by the much larger infrastructural challenges [62]. This image is supported by local data: Karim et al. discovered that rural students were about twice as likely to experience internet issues compared to urban students [56], and Hasan et al. have shown that poor technological availability was strongly linked to e-learning stress, which was disproportionately high in female and rural students [13]. Saarmean et al. have indicated that online learning is generally a source of dissatisfaction among students of private universities [16], and despite the generally positive attitudes towards Google Classroom, students still reported connectivity, interaction, and assessment-integrity issues [15]. Current research, though, is more likely to consider students or faculty alone, a single platform, and only infrequently integrate qualitative richness with quantitative validation or apply their results to context-based design solutions [27]. The current research fills this gap by using a mixed-methodology to determine the usability issues that are compromising the effectiveness of LMS in the Bangladeshi setting, as well as to guide a prototype that is responsive to local demands.

1.2 Rational Of The Study Or Motivation

LMSs are now a necessary component of higher education in the world but their performance remains extremely flawed and invariably accompanied by mistakes and challenges in usability and which negatively affects significant use. This is especially important in the setting where the scarcity of resources exists, such as in our country of Bangladesh, where the natural shortcomings of LMS platforms are put in contrast with the physical shortcomings, such as inconsistent internet connectivity, lack of access to devices, and disparities in the level of digital literacy. The institutional investments in these systems are usually high in form of file repositories unlike interactive learning systems. This translates to wasting an enormous amount of learning resources and losing the chance of improving learning. Most of the literature available is founded on the Western-centered framework of evaluation models that do not take into consideration the particularities of pedagogical, sociocultural, and infrastructural possibilities of the developing nations. This gives an acute lack of understanding of both how the barriers to usability nuclear to the institutions of higher learning in Bangladesh specifically are and what solution context-specifically might transform the LMS image into the realm of underutilised tools to successful learning ecosystems. Given the increased pace of digitalization of education after COVID-19 and the fast-growing market of e-learning in Asia, these usability issues have ceased to be a minor inconvenience and have become a key to equitable, high-quality education and to the full utilization of technological investments in institutions.

1.3 Problem Statement

LMS underutilization and usability weaknesses in higher education in Bangladesh.

Despite the popularity of LMS platforms in our universities, they are under-applied and in most instances, they are used as file repositories rather than interactive learning platforms [62][56]. Poor usability issues like unintuitive navigation, crowded layout, lack of mobile optimization, ineffective collaboration tools, and ineffective interface design do not encourage teachers and students to engage meaningfully with them [57][62][55][54]. Teachers often complain about the lack of training and institutional support, whereas students mention the lack of such features as search functions, feedback mechanisms, and interactive tools [62][56]. These constraints lead to frustration, disengagement, and resistance to adoption, particularly in environments such as Bangladesh where infrastructural challenges (internet reliability, device access, digital literacy) exacerbate usability shortcomings [56].

1.4 Objectives

We primarily aim to have a more convenient and effective LMS system within higher learning institutions. To do so, we have the following objectives we are looking forward to -

- To determine and examine usability issues that are paramount in the eyes of LMSs, including interface design, navigation, ease of access and teamwork fea-

tures among the students and the faculty members. We are planning to achieve that through our qualitative and quantitative approaches. By conducting all our analysis parts we want to find which way our participants are more keen towards. We are planning to extract the data if unappealing user interfaces, excessive operational steps, cluttered layouts - disturbs our participants productivity or not. Additionally, is there a communication gap between the students and faculty members just because of ineffective collaboration tools.

- To examine the relationship between the usability deficits and the impact on the LMS adoption rates, learner engagement, and the teaching methods in the institution of higher learning. We are planning to achieve this through our quantitative analysis part. We want to observe the participants preference through our study and then which one of our hypotheses are being expected by the statistical analysis. If our hypotheses get accepted then it will mean we need to actually work on this and it is impactful in the process of higher studies. Therefore, rejection of null hypotheses would mean we actually have to rethink our whole usability constraints and redesign it in a student friendly manner.
- To assess cases of the impact of institutional support mechanisms, professionalism development programs, and digital literacy skills on successful LMS use with a particular educational context, where resources are limited. By doing this we are trying to understand that to what extent a user has dependency on their institutional support when it comes to LMS usage. Also, if the participants believe that their faculties need professional development in this sector. If our quantitative analysis shows the dependency is not on the institution but on the participants' own effort, then this is an upgradable area. Because digital literacy should be implemented even if there are resource constraints.
- To produce evidence-based recommendations and implementation guidelines for designing and implementing changes that can enable institutions to enhance the usability of LMS and to design interactive and collaborative learning environments. With the help of our qualitative and quantitative analysis, we will be able to highlight the lackings which are the workable fields in the LMS adoption sector. These outcomes can help with enabling institutions to reduce the fragmenting tools and do their work within one singular core LMS.

1.5 Methodology In Brief

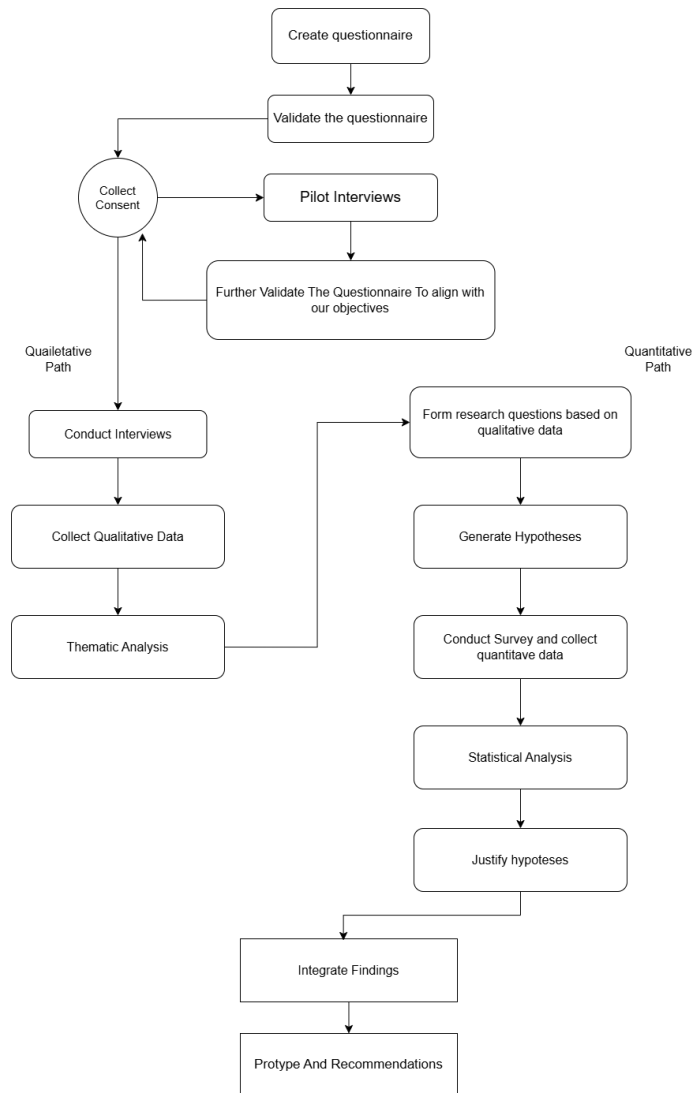


Figure 1.1: The flow chart of the workplan

In order to conduct our research, we followed the sequential mixed-methods approach to evaluate LMS usability. Firstly, we created a questionnaire which included quantitative and qualitative data. After creating the questionnaire, we checked if the questions are relevant towards the goal we wanted to reach. Thirdly, with the help of pilot questions and initial questions, the newly created questions were modified accordingly. After that, the modified questionnaire was released for actual data collection and then thematic analysis was organized on the collected data. On the other hand, with the help of qualitative themes our research hypotheses were developed. These hypotheses were validated through a quantitative survey, to measure the severity of the identified usability issues. After the quantitative analysis, we proceed to analyze the results and merge the results. Lastly, the results were concluded with future recommendations and a design prototype was created which will work as a recommendation to mitigate the observed problems. The Figure 1.1 provides a high level view of our proposed work follow ups.

1.6 Scopes And Challenges

Our study primarily focuses on usability and user experience evaluations of Learning Management Systems within Bangladeshi higher education institutions, examining interface design, navigation, accessibility, and collaborative functionalities from both student and faculty perspectives. Therefore, this study is a mixture of technological, pedagogical and sociocultural factors and we are examining the digital systems through the lens of a two-fold mindset for better understanding. This study consciously frames its evaluation of user interface design, navigation paths, and collaborative tools to understand how a participant's daily academic activities are influenced by the platform characteristics. Additionally, with the help of in depth interviews this study captures the unmapped and subtle barriers in higher education. Alongside, the quantitative analysis provides a scope while ensuring holistic exploration of platform usage by the users. Moreover, the diversity of students adds more intensity to this study because various opinions within a similar group have been observed. Altogether, these scopes provide multidimensional evaluation while keeping it contextually grounded and this consciously contributes to LMS effectiveness in resource constrained counties like ours. As this study employs mixed-methods

analysis combining quantitative surveys and qualitative interviews we have faced multiple challenges. As the majority of the participants were collected through peer networking, this study is more prone to selection bias. This methodological challenge can make the interviewers reach to the more tech-familiar and opinionated participants. Furthermore, social desirability effects and the recollection bias can be introduced by self-reported data of the participants. In addition, the study could be limited due to the sampling, whereby the sample size may be small and hence the findings may not be generalized to the diversified higher education context of Bangladesh.

We had to go through data collection challenges due to the uneven technical literacy. Sometimes we had to explain LMS concepts in the mid interview which compromised the depth of it. Moreover, not all participants felt comfortable disclosing their institutional LMS shortcomings though they were assured with anonymity. This behaviour to some extent contributed as a barrier in predicting constraints of the actual scenario. Furthermore, as the researchers are from the BRAC University so the data collection is heavily focused on their own university. This may lessen the path of skewing the early findings. In addition, we faced not so well structured responses when we were collecting the quantitative data and it can be said this was a result of not being aware of the actual LMS usage.

This study does not correlate with technical backend performance, server infrastructure, or cybersecurity aspects of LMS platforms. In future this can be an interesting sector to incorporate prototypes. Furthermore, there are no experimental interventions to test the usability improvement and ability to track user behavior over a long period of time. Integrating this tracker could be used to make the prototype more user friendly. In addition, the study concentrates on the direct instructional and learning experience instead of the administrative perspectives, learning analytics integration, and the mobile-specific LMS experiences. In cooperating these

perspectives in the study can make the intensity of the prototype more accepted. Also, instead of institutional assessment, user perceptions can be applied in exploring the contextual factors such as institutional policies, technical support structures and digital literacy training programs.

We have also faced a gap in formal LMS training across our participants. Due to this absence, some of our participants faced study barriers in the full adoption lifecycle of LMS. Moreover, we have found faculty inconsistency in platform usage contributes to highly variable student experience which is hard to generalize. Therefore, each student participants had various experience with the same LMS just because of this.

Altogether these challenges we faced due to the mixed-methods nature of the study. Though they are highlighted as boundaries, these can also serve as a gap fields to incorporate proper LMS usage in higher education.

Chapter 2

Literature Review

2.1 Preliminaries

Engagement in learning and learning outcomes are directly influenced by Learning Management Systems(LMS). The influence of LMS can be calculated by usability, accessibility, user experience. Combining technological efficiency with academic and sociocultural requirements is the solution to this problem. Though System Usability Scale(SUS) and User Experience Questionnaire are the conventional processes that are still widely implemented to describe user perceptions [62], some researchers still claim that these processes do not take into account vital educational and contextual factors [59].

For marking these loopholes, authors implemented approaches like- Information Systems Success Model(ISSM) and Technology Acceptance Model(TAM). These models put importance on the effect of perceived utility, quality and usability of the system that is adopted [24][54].Some also extend this approach by combining mixed-methods designs into think-aloud and eye-tracking protocols. This gives more insight into the external usability and wider learning process[25][56]. Although there remains some implementation and ethical concerns, learning analytics and visualization shows that use of basic data can be converted into real life development [60]. With the models stated before by the authors models emphasize the impact of perceived utility, usability and quality of the system in adoption [54][24]. Others also carry this practice further and incorporate mixed-methods designs into think-aloud and eye-tracking protocols, and this allows further insight into the superficial usability and into the bigger learning process [25][56]. Although there are still implementation and ethical issues, the application of learning analytics and visualization illustrates how unprocessed usage data could be turned into tangible changes [60].

Even though methodological diversity exists, most of the studies are cross-sectional snapshots, single-institutional information or self-reports on perceptions. Additionally, these have not been triangulated, with no longitudinal or multi-stakeholder effects of LMS improvements being known [28][21]. These arrangements emphasize the importance of additional and more flexible models to quantify and enhance LMS usability and user experience. And in a low resource country like Bangladesh we need to emphasize more on the operations that people have on a daily basis for academic purposes.

2.2 Related Work

The studies of usability and user experience of Learning Management Systems (LMS) have been increasing at a high pace in recent times, especially after the sudden shift to online education during the COVID-19 pandemic. Though this body of work has contributed to our knowledge of the interactions between learners and instructors with digital learning platforms, it is still divided in terms of the instruments used to evaluate usability, dimensions of experience deemed valuable to measure, and contexts of adoption that are examined. We organize the prior literature into three themes. To begin with, we examine the literature that assesses the usability and user experience of LMS based on standardized tools and evaluation techniques. Second, we discuss an emerging body of research that holds that the usability of interfaces, as such, is not enough, but a pedagogical, design, and sociocultural perspective should be taken into consideration. Third, we review the literature on LMS implementation in low-resource and Bangladeshi settings, where infrastructural and literacy challenges redefine usability as a practice. These themes combined drive the current work and identify the gap that they fill.

2.2.1 User Experience of LMS

Much of LMS research has focused on the measurement of usability and user experience using validated, reusable measures. The System Usability Scale (SUS) and the User Experience Questionnaire (UEQ) are the most repeatable with their benefits of comparability between systems and populations. The SUS to Moodle was tested on Hungarian university students by Berenyi et al. [52], who indicated that the mean score on SUS was 85.04, with the users who were the most satisfied with the quiz and the login functionalities, and dissatisfied with the notification tools and collaboration tools. In a mixed-method investigation of faculty, Koca and Özcan [57] used a mix of the SUS and nine in-depth interviews, which yielded a more moderate score of 63.85 and explained the discrepancy by inadequate infrastructure, ineffective design, insufficient support, and dull interaction tools. Using the UEQ, Ibrahim and Aziz [21] tested the UFuture platform on fifty postgraduate students and observed widely neutral ratings which were below the benchmark on the dimension of perspicuity and the lowest band on the other dimensions but there was a high level of agreement between users and experts. Saleh et al. [23] conducted a survey of 867 users and seven UX experts and reported that the strongest dimensions were perspicuity and efficiency, the weakest were dependability and novelty with a high level of agreement between users and experts.

In addition to questionnaire scales, researchers have scaled the acceptance models and inspection methods to the task of evaluation. Revyathi and Tselios [4] added to Technology Acceptance Model the SUS and through the Partial Least Squares analysis of 345 students, the self-efficacy, system access, social norms and perceived ease of use were identified as significant predictors of behavioral intention. Surface defects that self-report could be missed have also been detected by inspection-based and instrumented methods. Alihosseini et al. [50] used a cognitive walkthrough with thirty students, twenty-nine professors, two NAVID system experts, and as-

sessed the effectiveness, efficiency, and satisfaction and localized the usability issues related to password recovery and course management. In a study of fifteen female users of Blackboard, Alkoblan and Al-Wadud [25] [50] used eye-tracking, verbal feedback, and a questionnaire and identified thirty-one usability defects, about three times more than the eleven to fifteen defects that single-method techniques revealed. Comparative and standards-based testing expands the field even more: Başaran and Khalleefah [5] compared five open-source systems with the ISO/IEC 9126 quality model and found Moodle to be the best in terms of reliability, usability, and efficiency, and Ilmananda and Affandi [55] applied the TUXEL 2.0 method to integrate the usability problem reports with experience variables, identifying ambiguous instructions, weak feedback, and poor navigation in spite of positive ratings of motivation and system support. Tools that have been established to assess usability in related fields also guide this work: Islam et al. [8] reported on the usability of mobile health apps in Bangladesh and showed how structured usability assessment is applicable across the application domains.

This measurement-based strand is unified in two systematic reviews. Synthesizing forty-six articles through PRISMA protocol, Secaira et al. [61] found ease of use and user satisfaction as the most commonly researched quality criteria and Lu, Schmidt, and Shin [59] critiqued the reliance of the field on general-purpose HCI tools and suggested a general scale of usability to learning technologies, covering technological, pedagogical, and sociocultural usability and high reliability. The second theme is prefigured by the latter critique: that quantifying the interface, even in the most rigorous ways, only captures the LMS experience in part.

2.2.2 Pedagogical, Design and Sociocultural Aspects

A second line of evidence states that interface-level usability is not merely the necessary but insufficient and that the usefulness of an LMS is determined by the effectiveness of its design in terms of teaching, learning, and the social context in which it is used. Faghih et al. [1] present it explicitly by arguing that user interface design is not only an aesthetic issue, but a pedagogical tool, and that interactions and learning performance are enhanced where interface choices are based on cognitive load, motivation and multimedia principles. In line with this perception, Al-Fraihat et al. [54] examined qualitative narratives of 103 Moodle students and identified seven themes that influence success and discovered that otherwise useful functionality was compromised by unappealing design and low-quality content. Dzandu and Tang [2] reformulate the issue as that of human information interaction demonstrating that lack of context at realistic, semantic, and syntactic levels generated information gaps despite a well-structured content in a survey of 160 e-learning participants and suggest a model that combines platform and human information functions.

Surveys with a mix of instructor and learner views show that underutilization is often a pedagogical and organizational issue, but not necessarily a technical issue. In a study of forty-one teachers and forty-one students, Simon et al. [62] discovered that the majority of them considered Moodle as the file repository at most, that no one in the group had been trained to use it, and that the teachers experienced greater

difficulty and student are the ones who complained about the cluttered interface and slow speed also the search function. Almusharraf [28] applied Kelly repertory grid on twenty students and discovered that reliability and perceived importance had the highest weight to influence the satisfaction, and the priorities changed as students got experience. On the design innovation level, Ameen et al. [51] suggest an HCI concept of gamified interactive learning with Minecraft in Bangladeshi education, and Islam [3] describes the application of an LMS to introduce collaborative language learning in the tertiary education, both of which suggest that platforms can actively influence pedagogy and not merely host it. Rahman and Sanjana [15] consider Google Classroom as a part of tertiary English-for-Academic-Purposes classes and place the adoption of LMS in the context of a particular teaching practice.

The systemic and ethical aspects of LMS design can be also discussed in this theme. Ngulube and Ncube [60] have examined forty-one articles in the field of learning analytics and warned that excessive technical complexity can obscure the educational processes taking place, and that there is a gap between the collection of analytics and their application to aid the pedagogical process. The article by Diefenbach, Stoll, and Ullrich [53] in their general HCI literature conducted a survey of forty techniques of anticipating negative ethical and societal impacts of HCI innovation, and offers a lightweight approach of bringing to light harms that are unnoticed, which is particularly pertinent as LMS platforms become more widely involved in assessment and surveillance. Lastly, Hasana et al. [27] suggest a quality-assurance model of LMS that is specifically education and equality-focused, which is an indication that the concept of quality should be conceptualized pedagogically and socially, rather than technically. Taken together these studies indicate that assessment structures that are limited to the interface can misrepresent the systems they are assessed, a point that gains particular relevance when resources are limited [53] [4].

2.2.3 Adoption problems in Low-Resource Settings

The third theme involves LMS adoption in which a lack of infrastructural, economic and literacy fundamentally re-configures user experience. There are a number of studies that model the determinants of adoption in such conditions. Sobhani et al. [63] combined the Technology-Organization-Environment model and the Technology Acceptance Model of intention to use technology in a cross-sectional survey of 385 Iranian agricultural students who studied at three universities, revealing that technological and environmental variables were found to be decisive, with organizational ones not. Annamalai et al. [11] used an extended TAM to 205 Malaysian distance-learning students and discovered resources availability and ease of use to be key to positive attitudes, with troubles in finding the content and the absence of an interactive process being problems. According to Maslov et al. [14], learners in a largely online Finnish program were found to be highly dependent on Moodle and appreciated intuitive functionality and gamification, but due to small sample size per institution, they could not generalize.

A group of researches records the interaction between scale and seniority and infrastructural constraints. A survey conducted by Abdul-Gafaar et al. [49] on 14,481

students at the University of Cape Coast revealed that, even with the increase in academic performance by seniority, perceived usefulness and ease of use decreased with increasing expectations, low digital literacy, low connectivity rates, high data rates, and inadequate technical support were reported to be the main barriers. Other deployments in South Asia have similar obstacles. In a study conducted by Alebeisat et al. [20], 64 percent of the students in a university in Jordan considered the quality of HCI to be the heart of e-learning, especially in terms of instructor interaction and resource access. Sabir Ali et al. [24] found the self-learn problems, family and teacher related problems, connection and financial problems, and lack of training to be the most prevalent problems faced by students at the University of Baltistan, highlighting the importance of providing training to stakeholders in remote areas.

In Bangladesh alone, there is an emerging literature that predicts the gap between learners with and without resources. A survey of 130 and an interview of 25 in urban and rural universities indicated that rural students experienced problems with the internet about twice as frequently as their urban counterparts, that not many students preferred online to face-to-face classes, and that most teachers believed they could not fairly evaluate online, said Karim et al. [56] [13]. A survey of 508 students in from 9 private universities by Saarmean et al. revealed that online learning was generally ineffective due to poor materials, impaired well-being, and a significant increase in the rate of academic dishonesty, with physical classes being preferred by most students[16]. Hasan et al. [13] studied the use and access to technology as a means of e-learning among Bangladeshi university students and measured the differences in access to technology into uneven participation, Siddique [17] reported the impact of online classes at the tertiary level during the COVID-era. The issues are also related to secondary and professional education: Adnin et al. [19] observed how high-school students adopt online learning in Bangladesh during the pandemic, and Nurunnabi et al. [38] examined how e-learning can be integrated in the first year of the MBBS programme, revealing a range of opportunities and challenges both to students and instructors and to policymakers.

Combined, this body of literature confirms that LMS platforms are strictly assessed but limited in scope, that interface-based measurement often fails to heed the pedagogical and sociocultural circumstances that define actual use, and that such failures have the greatest impact in low-resource contexts like Bangladesh. The existing literature leans toward depending on individual institutions, cross-sectional snapshots and the viewpoint of a single group of users, and rarely tracks the subsequent solution of identified issues. The underlying reasons of the underutilization such as insufficient training, inconsistent design standards, and culturally unique barriers to adoption are not well understood. The current research is a reaction to this gap as it analyses the LMS usefulness on the Bangladeshi higher-education context using a multidimensional approach that incorporates interface, the pedagogy, and the infrastructural realities where the learning occurs.

2.3 Summary Of Key Findings

The majority of the past research on Learning Management Systems (LMS) have affirmed that usability, accessibility, and user experience are major factors that drive adoption and usage. They are typically based on the standardized measurement scales, including the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ) [23], or theoretical frameworks, including the Technology Acceptance Model (TAM) and Information Systems Success Model (ISSM) [54][24]. Although they have contributed to a deeper comprehension of user satisfaction and acceptance, they primarily focus on the superficial technological usability, and ignore the multifaceted interaction of pedagogical, cultural and infrastructural reality, in particular, in low-resource learning environments. The majority of studies focus on a single institution, cross-sectional data within a limited time frame, or one side of the research (students or teachers) only [28][62]. Consequently, the underlying problems of under utilization, including a lack of appropriate training, inconsistent design standards, and cultural barriers to adoption, are not well comprehended and are not adequately addressed. While standardized tools such as SUS and UEQ have analyzed comparable usability scores across platforms. They show limited to surface level evaluation of the interfaces and do not take into account the broader educational conditions that shape user experience [59][61]. With further indications showing LMS under-utilization is more often than not is the root cause of pedagogical misalignment and organizational factors. Which showcases that instructor efficiency over LMS's and interface design principle plays significant role in these. Also sociocultural context playing a decisive role in whether platforms are meaningfully adopted [62][1]. Coming to previously stated facts in countries like Bangladesh which have very limited resources are weighted down more by these infrastructural barriers with inconsistent institutional support significantly redefines what usability means in practice [56][16].”

These gaps are answered directly in the work when the mixed-method usability testing focuses on Bangladeshi higher education where there are infrastructural issues and the digital literacy issue makes the barriers to usability more extreme. The quantification of the study combines both quantitative surveys and qualitative knowledge to transcend the available superficial research to obtain both the quantifiable and experiential facets of the use of LMS. It accepts the different stakeholder perspectives namely, student, faculty and institutional factors to identify the hidden weaknesses of usability and the sociocultural constraints that stifle efficient interaction. Moreover, it aims at developing evidence-based design and implementation principles that are resource-constrained to enable LMS platforms to become learning environments where all learners engage in interactive and inclusive learning and where the learning is pedagogically aligned. This, therefore, not only adds to the existing discussion on usability, but also offers the world a localized, multidimensional theoretical framework to inform further system design, policy and training programs in Bangladesh and other developing environments.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

Research Design

- The study employed a mixed-methods approach combining:
 - Qualitative data collection through semi-structured interviews.
 - Quantitative analysis using chi-square tests and t-tests.
- 3 qualitative research questions formed the basis of the interview questionnaire, designed to elicit key thematic topics.
- 7 quantitative hypotheses were derived from qualitative findings to statistically justify and validate the identified themes.

Prototype Based on Key Problem Areas Addressed

- **Persistent Navigation Bar**
 - Consistent navigation across all platform
 - Public and private course hub
 - Global search functionality
 - Unified dashboard
- **UI & Usability**
 - Scoped materials tab for organized content access
 - Inline file preview card
 - Light and dark theme toggle
 - Accessibility toggles
- **Notifications & Communication**
 - Priority notification center

- Contextual notification preferences
- Color coded deadline reminders
- Integrated course discussion channels
- **Developmental Work**
 - In-app video tutorials menu and app support to address training deficiencies
 - Direct communication channel with the developer
- **Instructor Proficiency**
 - Role-based login system (Student / Teacher / Guest)
 - Faculty dashboard for assignments, quizzes, grades, and announcements
- **Session Management**
 - On-screen indicator alerting users when their session has been terminated and re-authentication is required
- **Motivation & Role Perception**
 - Menu-based quick access to advising, submissions, grades, quizzes, and communication

3.2 Societal Impact

Learning styles have changed greatly within the past few years, and with the onset of COVID-19, nearly all education shifted to remote learning. The LMS is the primary platform for student-teacher interaction. In Bangladesh, this has always been problematic. Our study aimed to understand these problems from the students' viewpoint, giving it a strong social relevance.

When we spoke to students from BRAC University, AIUB, North South University, and other institutions, one issue was common; they felt the experience was unequal. A student having a good laptop, fast internet, and a tech-savvy teacher had quite a different experience than one with an old device or a teacher uncomfortable with the platform- and that simply highlights the myth of technology as a great leveler once for all. In fact, it has only added another divide. If we can show this divide clearly and suggest ways to narrow it, the research can indeed support fairness in education.

When starting to use digital tools in university work, many students have to learn new tools alone. Due to this, there are some students who are less tech-savvy and are struggling with learning. Learning could be a lot simpler for many students if universities would provide better guidance and help. When I elicited information from students about how they learned to use new digital tools, the majority of students reported that they just started using the tool and then they worked it out. Some of them had to stop the machine if it had an error, and they had used YouTube

to learn how the tool worked. Some sought advice from their own friends who knew the tool well enough. This demonstrates that for many students the only support is from themselves, online videos or friends, rather than their University.

Improving the learning management systems in universities and providing proper training to teachers will make the digital learning better in universities in Bangladesh. The teachers are very important in assisting students to learn online. Students should also receive support to make them comfortable with the use of digital tools at the same time. We also observed that some students have a financial problem in our study. For instance, one student noted that a \$15 course fee in Coursera is costly for student in Bangladesh. While some learning materials are free initially, students may be required to pay later on. This has a disproportionate effect on certain students and makes it less likely that they will have the same opportunities as other learners. The use of the platform is not only a problem but a problem of fairness.

Our study does not list all the difficulties that students are likely to encounter but it does give a sense of some important issues for universities to be aware of. It's by identifying these challenges that we hope that universities and platform designers will gain greater insight into the challenges students face. If they are aware of student experiences, they will make informed choices. These decisions must be made based on the true experiences of students in their learning path.

3.3 Environmental Impact

The most obvious relationship that was identified was that between digital learning and paper use. A good Learning Management System (LMS) would allow access to the students of lecture slides, assignments, and feedback in a digital format. Therefore, notes are not required to be written on paper and hard copies are not required to be submitted. Many students point out this as a major benefit of Google Classroom and BUX, since all materials are in a digital format. This indicates that redesigning LMS systems can be a way to enhance the learning experience and minimize paper consumption in educational settings.

This is an energy angle, too. Several students commented that some platforms were slow and clunky and used a lot of resources. Discord and Microsoft Teams were the most common. Some in the discussion noted that these applications consume an excessive amount of RAM and consume too much CPU power. Unoptimized applications consume more power. This can make a difference, particularly when applied on a large scale over time. The lighter the platform, the easier it will be to use and is also energy-efficient.

There are also resources used by the digital infrastructure for each file uploaded, video stream, and each notification sent. Students could be spending all day on BUX, Google Classroom, Discord, Slack, and many other applications. This will contribute to increase in data traffic, server requests and also will consume more power than a single integrated system. The participants themselves expressed a desire to have a more streamlined LMS, which could help to eliminate unnecessary

digital overhead.

Our study is rather small and immediate in greening the impact. The construction of large infrastructure. We do not have scale service. Our work is to indicate the path to superior design and institutional choices. All it takes is minor modifications to cleaner code, some of the lightest applications, reduced platform switching and paper consumption. They may build up with time.

3.4 Ethical Issues

This research was approved by the local Institutional Review Board (IRB). All participants were undergraduate, graduate students and respected faculty members, so all of them are adults and after they gave consent for the audio recording for future analysis the interviews were conducted. Prior to each interview, the purpose of the research and how the data was going to be used was explained in detail. No force was used and the right of the participants to abandon the study any time without any explanation was honored. They also agreed to tape the sessions, which were necessary to be transcribed and analyzed further. In this study, anonymity was a huge issue. The participants were authentic students from authentic universities. Feedback contained certain criticisms of teachers or schools. These comments may be embarrassing or have other consequences if they are attributed to specific people. Because of this, the names were substituted with participant codes such as J1, A3 and T6. The reference to universities was done in a broad way so as not to create the impression that it was associated with certain criticism.

Also, there is an ethical issue on how findings are presented. Since qualitative research is about interpretation, there is a fear of presenting only quotes that back up one story. This risk was reduced by sharing both good experiences and problems that were noted. The codebook was also used to improve how orderly our analysis was. One more thing that is really important is the power relationships in interviews. The people doing the research and the people being interviewed often have backgrounds, which can make the people being interviewed want to agree with the interviewer or say what they think the interviewer wants to hear. To avoid this we asked questions that were open and did not suggest an answer so the people being interviewed would know that there are no right or wrong answers.

When we do user testing for the prototype we will follow the same rules. These rules include making sure the people being interviewed know what they are getting into keeping them anonymous and letting them stop the interview at any time if they want to. Showing someone an interface and asking for their opinion is not usually a risk. However it is very important that the people being interviewed understand what is going on and are free to say what they think including saying what they do not like about the interface. The people being interviewed should be able to give their feedback including suggestions, for how to make the interface better.

Our research holds a broader ethical drive. We are not into the making of a commercial product, neither are we collecting data for profit. Our main objective is to generate knowledge that can help improve learning experiences for students in

similar contexts. This purpose has driven every aspect of the study, starting from the selection of interview subjects to the methodology used in data analysis and the recommendations.

3.5 Project Management Plan

The results of the research were used to develop the prototype. The prototype facilitated an understanding of the research. This section is about how we managed the project what the prototype is and how the project and the prototype worked together.

The prototype is a complete functioning LMS application. It was designed as a High Fidelity System Implementation. Its main aim is to address usability problems in the findings of our mixed-methods study. The system has a partial database backend. It facilitates user authentication, role-based access control, material storage, assignment submission and graded feedback. Core functions are the same for both the student and faculty interfaces. The primary weakness is that it does not have live institutional data integration. Formal institutional access and API backend deals would be needed to connect the system to an active university course database. It is not a limitation in scope of this research, but rather a limitation in structure. It doesn't diminish the usability evaluation or design contribution of the work. An actual prototype was created instead of a visual mockup, as it would give the audience a clear idea that the design ideas in this paper are not hypothetical. Anyone who is reading this research can open the application and get a sense of how a better designed LMS could feel.

All features in the prototype were adapted from the interviews or survey. The design was not based on speculation of what an LMS should do but rather on the aspects where students and faculty claimed that it was either missing or not functioning in the systems they were using. The global search function is available across courses, materials, messages and announcements. Materials were repeatedly cited as being difficult to locate. They also reported that there were no helpful search features on their platforms. This was one of the initial features we focused on. Redesigned notification system to provide more control. This includes a notification center, priority tabs and preferences which can be set by the user. Interviews and survey results revealed that there was in fact a problem of notification overload. Many students went to the extent of muting channels completely. Therefore they missed some important updates. The dashboard gathers deadlines, announcements, course access and quick actions in one place. This is a direct reaction to platform fragmentation. Students reported going to four to five different places to find out what was due. The participants were definitely requesting consolidation. The dashboard is our answer to that need. Materials page features filters, search by course, recently viewed items, pin to and unpin from list, and preview modal. All these address the amount of time students have to scroll through unorganised file lists. Students told the developers that files were frequently getting lost in quick-moving channels in Discord. That's where the idea of the pin came from.

The prototype is basically pages for course management, announcements, materials,

assignments, and grading for faculty. Faculty can make materials publicly or privately available. This separation is important because the same content might be specific to the institution and others might be able to be disseminated more widely. Most current platforms do not make this distinction very clear in their management. For clarity, the assignment flow should contain a countdown to the deadline, display a rubric, allow file uploads, show confirmation modal, and have a submission banner. These were meant to clear up any confusion. One student commented that students might miss the final submit button in Google Classroom. In one instance, this led to an assignment being marked as not submitted when it was actually uploaded. The discussion system should allow the use of channels, pinned messages, threaded replies, mentions, and search. This feature was influenced by what students liked about Discord, which was mainly the organization of channels and the possibility of communicating directly. Meanwhile, they were not fond of the noise, the loss of the messages, and the absence of an educational system. The monthly and list forms are both deadlined. Deadlines are made interactive to solve the problem of fragmented deadline information on different platforms. Read receipts are added to the messages inbox to prioritize faculty contacts as the confusion over whether faculty have read a message is currently unresolved.

Our research revealed that there are students who are more impacted by usability barriers than others. Technology is a challenge for students with devices and students who are not technologically inclined. We also considered pupils with disabilities and pupils who become easily confused. The new system includes a link that allows you to navigate to the main portion of it when using the keyboard and special labels to assist screen readers. It also offers features for easier viewing and audibility such as a non-contrast colour feature to reduce the impact of colours, a button to prevent the objects moving around and a special font to make it easy for dyslexic users to read. Each color with a significance also has words to explain it so as not to leave behind those who don't like colours. We made these choices because we want to follow the rules that ensure everything is accessible to everyone and because we want to ensure our design includes all students regardless of their abilities. This is very important, for schools, where all students should be able to use the learning tools no matter what their abilities are.

The prototype features a light/dark toggle and offers a nice, compact density switch. This allows users to determine the amount of information they wish to display on the screen. There's a partial English and Bangla language toggle as well. This is important because Bangladeshi population is the research population. Accessibility of the language is a genuine problem. The majority of LMS used in Bangladeshi Universities are English only. This can be a deterrent for some people. Role switching enables the user to switch from the student perspective to the faculty perspective. This was included partly for demonstration, partly to show its applicability to real academic situations, as some users may assume both roles. Part-time and GTA teachers can be enrolled as students in the system. The difference between public and private content is that the public can access the course section without sign-up, which is referred to as the Guest Browsing Mode. Full access to institution-specific materials can only be acquired after verification from a university email address, which provides an appropriate level of access control.

3.6 Risk Management

This study, like any other research, has its own set of risks. Some factors can restrict the results and others can restrict the use of the results. The biggest risk is that of the sample, which consisted of thirty students and faculty members of private universities in Dhaka. What happens at public universities can differ from student to student. There can also be other problems with those who come from rural institutions. There can be significant differences in results, depending on the institution's resources. So, it should not be generalized about the use of LMS in Bangladesh. The application of our findings can be more confidently made to private, urban, English medium institutions. Another problem is honesty during an interview. We assured them of anonymity. Nonetheless, the responses an individual can provide may be suitable, but not the answer one might hope to hear. A few respondents may not have complained about their institutions or teachers. This is a known problem in interview-based research. It implies that the results are perhaps not quite as bad as they seem. The other risk is codebook. It was developed meticulously, but still, it might not capture all. Interpretive coding is qualitative coding. Different coders can give the same quote different codes. This was minimised by having several raters and calculating inter rater reliability. However, there is some subjectivity inherent. Other studies may have recognized slightly different themes.

The major risk on the prototype side is that of scope creep. There is also a difference between what the software can show and what would be required for a deployed LMS. Prototype is an actual running application. It does not have a connection, to a live university backend. This implies that when the participants conduct the user testing it must be carried out in an environment. The participants will understand that they are NOT working with a system in this setting. This can impact the way the participants interact with the university application naturally. When conducting the user testing, consider not using the university application in this way. The feedback should thus be taken as a reaction to a demonstration and not as being from a full field trial. Finally, LMS software is constantly evolving. A few specific issues raised by students may already be somewhat resolved, such as Canvas logging people out a lot or Discord' losing files. Our research is about one moment in time and not a single snapshot of an existing situation. This is standard for studies of this nature, but it's important to keep in mind.

Additionally, as the prototype we built works with both private and public academic contents, there is also risk of private academical content volatilization. Anybody can act maliciously toward the private academical content and thus the we are determined to work on this concerning aspect of our prototype in our future work.

3.7 Economic Analysis

The relationship of economics in this study is not with the profit or business models but with the cost in a wider sense. When learning technology is inadequate, it has a cost to students, institutions and society. Having a better LMS design can lower some of those costs.

The economic fact that stands out is the affordability. Students are already on a combination of paid and free platforms. Some of these are also shifting to a paid service. One participant provided a good example. A couple of years ago, platforms such as Coursera provided financial aid and free access to the certificate. Now some require monthly subscriptions. Ten dollars a month or fifteen dollars a month is a substantial amount of money for a Bangladeshi student. If LMS systems are to keep on going down this route, some learners will be left without access to quality education. Indirect costs exist, too. Perhaps, the student requires to pass through five or six platforms in order to accomplish his academic work. There is no learning time in all these: finding materials, logging in and out of systems, muting notification channels, searching for lost files, etc. Real cost has this friction but it does not show up on a financial statement. Institutions incur the costs as well when their investments in LMS do not yield desired outcomes. Universities pay for these, such as Canvas, BUX and Microsoft Teams. If these are used merely as a file storage these are not so beneficial in terms of return on investment. Our research shows that there is a correlation between limited use and usability and training problems. And such issues can be addressed without additional big spending.

The prototype is not a product and is not present in the market. What we're saying is that it's not ready to be an alternative to an existing LMS. However, it is an effective application. That's important for the business case. Global search, role-based access, notifications, assignment workflows, discussion tools, and materials filters, among others, are included in the prototype. It is also associated with a database. This shows that these enhancements can be made using average resources. All tools used in the construction of the whole system were free and open. This was intentionally done. One of the main outcomes of our work is that the students do not ask impossible engineering solutions; they ask to design better and make clearer development decisions. That better LMS experience may not be as costly as they believe it to be to make. The fundamental work has been done in case any university would like to use or alter our design. Main costs would be server infrastructure, institutional data integration and faculty onboarding. The school would not have to reinvent itself with regard to design and development.

The four years of poor learning material is time wasted. Improved LMS experiences can contribute to improved digital fluency. And that is good to employers, the tech industry and the economy at large. We are not saying that LMS usability alone can change the economy. It's one small part of a much larger issue. Education systems must adjust to the digital reality. Correctly managing this aspect is more than beneficial in the university setting.

Chapter 4

Proposed Methodology

4.1 Methodology

Our study was conducted over a 4 months period of time in Dhaka, Bangladesh. The interviews for our study were taken physically and online. Following proper consent forms and semi-structured questions. For the consent process, the physical interviews were easy as they signed the physical copy of the consent form which gave us the consent to conduct the interview record and transcribe them. The online interviews were quite different as we collected both verbal and google form based consents for each participant so that we can ensure their consent thoroughly before conducting interviews. The interviews lasted between 10-20 minutes based on the individual. It was mostly taken from university students and faculties but some of the participants also were recently graduated students. And our qualitative interview was done based on convenience sampling as we were university students ourselves it was easy to get hold of students and faculties in our close quarters. Adding to that, there were several participants who were from other universities. We knew them as they were already our peers from other universities so we approached them with the interview invite as they fitted our criteria without any specification. As we wanted to approach this problem through a exploratory sequential mixed-methods perspective. After conducting a number of interviews we formed a survey type questionnaire for quantitative data collection based on the findings we got in our Qualitative interviews. This would help us justify the severity of these problems and show how these findings are making the LMS field more and more redundant from the perspective of our country.

The semi structured interview questionnaire was designed using the questionnaire format that included open-ended questions which were divided into four thematic areas entailing: (1) usability concerns such as interface design, navigation, ease of access, and the ability to collaborate with other users; (2) effect on LMS adoption that focused on relationships between usability deficits and adoption rates, learner engagement, and instructional practices; (3) support provided by institutions assessing support mechanisms, professional development programs, and digital literacy programs; and (4) evidence based recommendations that included participant suggestions of how to improve LMS use. The structure made sure that every interview that we took addressed the objectives that aligned our study. And this also allowed participants to elaborate on their experiences.

The target population we wanted for our study consisted of university students and faculty members who actively used learning management systems for academic purposes. Participants also included of graduate students who use and used LMS systems for their academic and research purposes. With the convenience sampling approach, which was facilitated by the peer based nature of the research setting. For this we and the participants shared similar academic communities. Initially for conducting interviews we took on participants from our own institution. And after which the sample was broadened to include participants from other universities. As a result, it helped us to get a wider range of perspectives and institutional backgrounds. For ensuring relevant and ethically sound data collection and specific inclusion criteria were established to make solid justification of our study. Participants were students and active faculty members at educational institutions. Which made them to be regular users of at least one LMS platform, and to be willing to participate in recorded interviews and provide informed consent.

Academic Status	Department / Discipline	No. of Participants
Lecturers	CSE	8
	Other Departments	2
	Subtotal	10
Graduate Students	CSE	3
	BBA	1
	Subtotal	4
Undergraduate Students	CSE	13
	BBA	6
	Bio-Technology	1
	Subtotal	20
Total Participants		34

Table 4.1: Participant Demographics

Our study started with 5 sample participants where 3 of them were graduate students and 2 of them were undergraduate students. which we conducted to see if our semi-structured interviews gave us the outcome we hoped for. Basically sample interviews that determined if the questionnaire outcomes aligned with our study's objectives. This served the purpose to get us more familiarized with the on field interviews and how they are perceiving problems in their regular LMS system. As we validated our questionnaire through the sample interviews. These interviews were conducted and recorded with participants consent maintaining anonymity for their identity. We conducted our study firstly based on asking their roles and what kind of platforms they use for conducting their academic activities on a daily basis. Which further along included the ease of use of the specific platforms and how they perceive the role of it or them in learning or teaching. Our questions also consisted of questions which asked them about their preference over any platforms and what caused them to be more reluctant towards LMS systems and if they wanted any improvements in the LMS's they use.

After this, we proceeded to the first phase of interviews, which consisted of 10 more participants, where all of them were undergraduate students. Through interviewing them we observed about the various aspects of LMS platforms that they use that causes them problems and hampers their workflow. It helped us to learn various aspects of LMS platforms that had been talked about all the time but never been addressed institutionally by any of the authorities to work on them more evidently to make the workflow through LMS more efficient and more stable.

In the next phase of our interviews, we conducted 19 interviews, where there were 10 lecturers and 9 more undergraduates. The inclusion of faculty members in this phase gave us very important aspects of LMS systems for our study, as their perspectives provided an institutional insight that complemented the student focused data that we had collected previously. Interview questions for lecturers were structured around the same four thematic areas. They also aligned with the contextual adaptation to reflect their role as instructors focusing on how they selected and managed LMS platforms, what features they prioritised, and how institutional directives shaped their platform choices. Through their data we noticed that it always came down to the specific course coordination and the comfortability of the instructor and the authorities direction of control over the course.

After the interviews were completed, we proceeded to the data analysis phase, which was conducted after the transcription of all recorded interviews. This process followed thematic analysis of the codes we extracted and moving from initial codes to code clusters and finally to overarching themes. Firstly, We carefully looked into each of the interviews and took out specific quotes that were addressing problems, concerns and impact because of LMS systems. After that for each of the quotations we took out an initial code formatted for the problem that was addressed through the participant's interviews. Conveniently, throughout 34 of the participants there were several problems which were addressed. So to get a more specific outcome for our study purposes we relied on the 4 raters. And based upon the rating of each quotation and initial code we got some problems that were recurring and causing actual problems. With the stated initial codes we created code clusters which helped us to extract themes for every one of the initial themes that were addressed initially. Hence, giving us the final findings of our study that addresses our objectives.

After the last phase of qualitative data collection and analysis, we had gathered sufficient data to identify and articulate the core problems of LMS systems within the context of our study. We needed a quantitative measure throughout several participants who had experience over using LMS systems. Participants varied from undergraduate students to job holders who recently completed graduation and were used to the hassles LMS systems. and their opinions over the problem to justify the themes that we addressed in our study. So through specific hypothesis which addressed each of our theme and mapped them with certain questions that addressed the problems that were discussed we carried out some research questions based on our themes created hypothesis and based on that we created 15 variable questions that would justify the severity of the problems that we addressed and help us to prove the causes that made the LMS redundant in our country. To test each of these hypotheses, a hypothesis pair was used, with the null hypothesis (H_0) and alternate

hypothesis (H_1). In each case, the null hypothesis claims that the variable being tested does not have a statistically significant impact on either student engagement or adoption of LMS. On the other hand, the alternative hypothesis, inspired by the results of the interviews, claims a directionality or distribution about student actions or perceptions. The null hypothesis and alternative hypothesis follow a logical pattern wherein the former is either rejected or confirmed based on its probability of occurrence.

All participants who took part in our research gave informed consent before the interviews and we clearly stated the purpose of the study, the way it would be conducted, the rights of the participants, how the data would be used and the option to withdraw at any given time without any ramifications. Any anonymity was strictly adhered to during the research procedure and no records or disclosure of the participant identity were in any reports or publication and this was especially important considering the fear of confidentiality of institutional information. The confidentiality measures stipulated that the recordings of the interviews and the transcripts were well stored and only we as a team could access the information and the data was presented using aggregate analysis anonymity regarding the individual and the institutions. We overcame this reluctance to talk about institutional support mechanisms by reinforcing the ethical commitments to anonymity and describing data related processes, which is essential and ensures ethical transparency to receive genuine responses. The research was conducted in accordance with institutional ethical principles in conducting research with human subjects and especially sensitive institutional data.

Our study also faced several methodological limitations as well. With respect to sampling, the reliance on peer networks and convenience sampling could have introduced selection bias in our interviews. Conveniently, over-representing participants with stronger opinions about LMS or those who have higher levels of technological familiarity. The initial emphasis on the home institution may also have skewed early data trends. Participants also had several technical literacy problems, as some required explanation of LMS related concepts. From a scope perspective, the qualitative methodology while rich in contextual detail carries inherently limited external validity, and the findings may not generalise to institutions with substantially different levels of resources and also how each academic institution structures their process.

4.1.1 Qualitative Dataset Demographics:

In our research thirty four participants from multiple universities were involved across two separate phases. Phase 2 mostly consisted of faculty interviews. Which represented a combined level of study from the students and the faculties. In different levels of higher education, variation in academic levels gave a perception of LMS usage. During this, a strong focus was given on learning experiences during undergraduate studies in which the relationship with the LMS was mandatory and frequent.

Several participants described having various LMS and learning platforms open at

the same time. This gave a reflection of a divided online learning ecosystem without dependency on a single centralized system. It was noticed that Google Classroom was the most commonly used platform, as reported by 22 participants. Then came discord from 17 participants and Slack from 11 participants. BUX type institutional LMS was reported by 7 participants. Microsoft Teams was reported by 1 student other learning platforms like Coursera and Udemy each. There was inclusion of Canvas of NSU and Zoom by 2 participants each. Also an administrative LMS came into play which Connect mostly contributed to administrative work. This distributed usage of LMS platforms reflects that the LMS usage of participants is not limited to formally given institution systems. It was noticed that many times students mixed institutional LMS platforms with communication-focused and other learning platforms. This pattern of using LMS platforms supports the aspect of platform diversity and fragmentation directly, in which participants give descriptions of going through several platforms for managing access to content, submissions, discussions, and feedback. The way different groups use platforms shows that fragmentation is not just a coincidence, it is built into the academic workflows of participants.

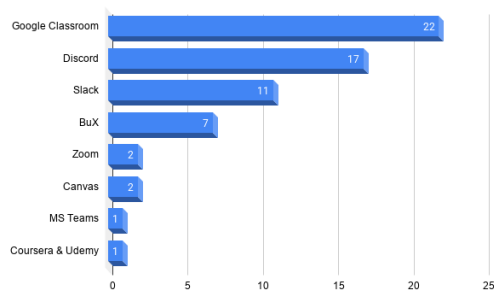


Figure 4.1: LMS Platform Usage Patterns- 1

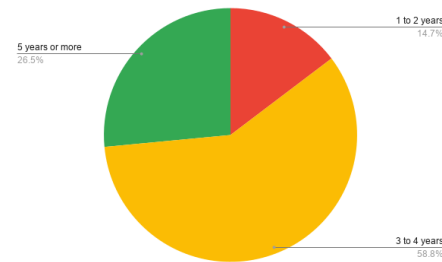


Figure 4.2: LMS Usage timeline

With regard to experience, most of the participants were long-term LMS users. About 58.8% respondents said that they have been using LMS platforms for 3 to 4 years and 26.5% had years of experience of 5 or more. And 14.7% had used LMS platforms within a period of 1 to 2 years. It means that the majority of the participants were heavily exposed to LMS settings, usually starting during or prior to the COVID-19 pandemic and carried on into post-pandemic academic existence. This prolonged period of use is especially relevant in understanding usability issues that were brought out in the findings. Although with years of experience, the participants still claimed confusion, navigation problems, performance problems, and not-organized contents, as addressed in themes like user interface and usability concerns also technicality challenges and infrastructure limitation issues. This is an indication that even experienced users continue to face usability problems that are not a consequence of not being well acquainted with them or adjustment issues. Rather, they are the issues of a larger design and implementation constraints of LMS platforms.

The demographic profile explains the patterns in learning curve and training deficiencies as well. Although the majority of participants claimed that the use of LMS facilitated with time, it was mostly due to self-education, trial and error, and

peer support, and not to institutional training. The fact that the participants are exposed to LMS over a long period of time adds to the conversation that LMS proficiency is formed in the absence of extensive onboarding, thus enhancing the fact that training and support is an unaddressed institutional issue. On the same note, we can notice role perception and motivation is put in context by the high LMS experience among the participants. Regarding participants, in spite of years of usage, many still viewed LMS sites as content repositories and not as active learning spaces. This means that long-term use does not necessarily result in greater engagement or motivation. Rather, significant interaction is based on the usability of platforms, effectiveness of communication, and engagement of the instructors. Generally, the demographic analysis shows that the usability issues that will be revealed in the present study are not unique to the inexperienced users or the particular platform. Instead, they are developed through long term, daily exposure to fragmented LMS ecosystems, which further support the main thesis of the study that there exist invisible obstacles to effective use of LMS systems at the design of the system and in institutional practices, and in long-term implementation strategies.

4.1.2 Quantitative Dataset Demographics:

For the quantitative study 222 participants answered to our survey. All of them were using LMS platform actively. We designed this survey to justify our qualitative data through hypotheses based process.

If we look at the academic positioning of the survey it composed majority of undergraduate students which consisted of 168 respondents which represented 75.7% of the total sample. After that most amount of LMS users that arose are postgraduate students with the number of 37 respondents which can be affiliated to 16.7%. Then 12 respondents were graduate students who had experience using LMS systems. And the other 5% were various other fields such as pyrotechnical students etc. With this variation of respondents it showed that the convenience sampling that was done for qualitative data it had chances of having skewed data.

Within our survey respondents were asked to indicate which LMS systems they used for communication and academic purposes. With multiple selection permitted. Email emerged as the most used one of course it's not a formal LMS system per say. But over 153 responses reported using it in their everyday academic works. After that came in Discord with 113 responses. Which signified it as a strong adoption specially being a informal tool from the perspective of academic institutions. Slack was reported by 92 responses. Reflecting its use across both peer collaboration and course-related communication. Then What'sApp and Messenger were reported by 72 responses further confirming reliability on consumer grade messaging platforms over dedicated LMS systems. Other than that, major LMS platforms such as Google Classroom were used by 13 responses. while other platforms such as Canvas, Canva, bUx, and Telegram each registered very low responses indicating usage within the surveyed population.

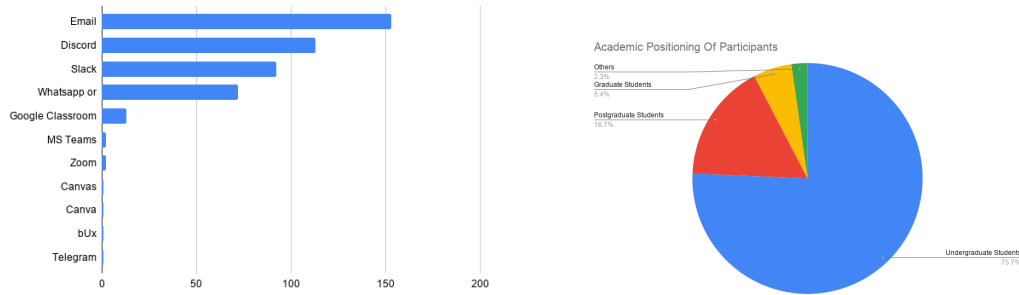


Figure 4.3: LMS Platform Usage Patterns-2

With the data combined the demographic profiling of the quantitative data showed that it was highly responded by undergraduate students who are digitally active across several platform because of study related communication and academic purposes. Hence, providing our study a representative basis for evaluating the severity of LMS usability deficiencies identified qualitatively. And with the breadth of participation across multiple institutional respondents it strengthens the generalizability of the quantitative findings for local context.

4.2 Design Specification

4.2.1 Qualitative Research Design

In order to carry forward with our qualitative data collection, we firstly generated some research questions which mapped our objectives and problem statement. As our problem statement deals with the under utilization of LMS and usability weaknesses in our country, we connected some research questions to it-

- **RQ1:** What are the contextual factors unique to our higher education environment that shape how students and faculty perceive and interact with LMS platforms?
- **RQ2:** How do students and faculty describe the role of the LMS in supporting collective and communicative dimensions of their academic experience?
- **RQ3:** What improvements do LMS users prioritize, and how do their suggested changes reflect their unmet academic and pedagogical needs?

Firstly, we aimed to observe the unique contextual factors which are shaping how students and faculties are perceiving and interacting with LMS platforms. As user observance and connectivity can make positive and negative impressions on the user, therefore knowing the root cause is important for our research. Secondly, we wanted to understand the impression of LMS on the student and faculty both sides, regarding collaborative and communicative dimensions in academia. This helped us understand the role of LMS from the dual perspective. Thirdly, we wanted to highlight their priorities regarding LMS which will be a reflection of their unmet academic and pedagogical needs. This worked as a pathway to know about the

improvements they wanted in a LMS for their academic benefits according to them. All these research questions motivated us to create our semi-structured interview questionnaire which helped us in collecting qualitative data. Each question helped us collect real time data from the participant's perspective with personal experience. While conducting each 15-20 minutes of interview we were successful to fulfill our research needs with our open ended questionnaire, which also allowed us to do prob questioning to understand more depth regarding our research.

Several measures were undertaken to ensure the validity and reliability of the study. For validating the content we established by closely aligning the interview questions with the research objectives and the dimensions of the theoretical framework. And with this the pilot testing confirmed us that the questions effectively drew out the type of information that we needed. To structure the reliability aspect more strongly we used a standardized semi-structured questionnaire across all interviews, which ensured consistency while it also gave flexibility to get necessary and meaningful dialogue. For the Credibility we supported it through sustained engagement with the research context and with the perspective of our teams five raters during the coding phase to reduce individual bias. Transferability was addressed through detailed description of our research's setting and profiles of the participants, which allowed readers to assess the applicability of the findings to different institutional contexts. Dependability was ensured by systematic documentation of all research procedures, including the modifications that were made during the pilot phase. Finally, all raw data that was collected during the phases are preserved in the form of recordings and transcripts. Ensuring conformability and allow for independent audit if required.

Building upon all of these findings and the severity of each theme we developed a prototype that addresses the usability problems identified throughout the study. The prototype we created is supported by both qualitative themes and quantitative validation. It incorporates design level implementations and functional suggestions based on the data we gathered. It also gives us perspective over how the recurring deficiencies in current LMS platforms can be resolved through targeted design interventions. These were accompanied by a set of contextually grounded recommendations which coherently will improve LMS adoption and effectiveness for both student and teachers ease of use.

4.2.2 Quantitative Research Design

We used two types of statistical tests, which were based on the format of the responses (question types) and the scale of measurement of each question used in the analysis. Chi-square goodness-of-fit tests were conducted for all categorical and ordinal (checkbox multi-select, single select dropdown, and ordinal frequency scales) questions. The most widely used and familiar method of deciding whether the distribution of the observation responses is significantly different from a uniform distribution (i.e., whether one or more response options has been overrepresented) is the chi-square procedure. This is appropriate when the research question is about the distribution of responses for example, do students group around certain responses, such as more people reporting on these interface issues than others, or do most students communicate with others outside the LMS instead of within the LMS. For

Likert-scale items rated on a 1-5 continuously-scored scale we used one-sample t-tests, comparing the population mean of 3 with our observed sample means. For each scale question a directional type of t-test was used, because this is the manner that is embedded within its hypothesis. Hypothesis tests for higher ratings than neutral were conducted using a right-tailed t-test on each of the anticipated questions (e.g., disruption from platform switching (H2), severity of technical failures (H3), instructor proficiency (H6)). To demonstrate that the motivation ratings of LMS are significantly lower than the neutral point of the LMS motivation scale, H7 was tested with a left-tailed t-test of $H \pm 1$ and below the neutral point of the motivation scale, using 0 as the neutral point.

A significance level of ($\alpha = 0.05$) was used throughout, which is typical in social science research and HCI. If one or more of the tests associated with a hypothesis has a p-value less than this, then the hypothesis is supported and we reject H_0 . The wording acknowledges that this minimum p-value is used to make the decision to reject a hypothesis with 2 questions as part of the instrument; it is a multi-indicator instrument. As each pair of questions assesses the same latent construct from different directions, a positive finding on either question is not just evidence for an effect of that question on the outcome; it's evidence for an effect of the construct itself. This corresponds to the logic used for construct validation: that is, if one indicator is significant, that indicator is evidence of statistical significance for that effect, although it may not be at threshold significance in the second independent test. This is in contrast to a situation where we would have to have everything pass, which would impose an unacceptably high bar that would be inconsistent with the design of the survey in multiple items, mixed format. The sample size of 222 was large enough that small effect sizes would be detected and the minimum p-value technique was selected as the theoretically correct decision making for this instrument, keeping the Type I error rate in check.

Hypothesis	Metric	Description
UI & Navigation	RQ1	To what extent do interface design issues reduce university students' engagement with their LMS?
	H ₀	LMS interface design does not significantly affect student engagement.
	H ₁	Poor UI design and navigation complexity significantly reduces student engagement with LMS platforms.
Platform Fragmentation	RQ2	How many digital platforms do students simultaneously use for academic purposes, and to what degree does switching between them disrupt their workflow?
	H ₀	Using multiple platforms simultaneously does not negatively affect the academic experience.
	H ₁	Students using 3 or more platforms simultaneously experience significantly higher academic friction and cognitive load.
Technical Infrastructure	RQ3	How frequently do technical interrupt academic tasks, and how severely do students perceive their impact?
	H ₀	Technical performance issues do not significantly impact LMS adoption rates among students.
	H ₁	Frequent technical failures significantly suppress LMS adoption.
Training & Institutional Support	RQ4	To what extent do our universities provide formal LMS training, and how does the absence of institutional support affect student's confidence and proficiency?
	H ₀	The absence of formal LMS training does not significantly affect students' ability to use the system effectively.
	H ₁	Students who received no formal training will report significantly lower LMS proficiency and engagement.
Notification & Communication	RQ5	Does LMS notification overload or failure significantly contribute to students abandoning platform-native communication tools in favour of external applications such as WhatsApp or Discord?
	H ₀	LMS notification systems do not significantly influence whether students migrate to external apps.
	H ₁	Students who experience frequent notification failures are significantly more likely to abandon LMS communication tools.
Instructor Proficiency	RQ6	How does instructor engagement and proficiency with LMS tools moderate student motivation, course structure clarity, and overall platform usage patterns?
	H ₀	Instructor proficiency with the LMS has no significant effect on student learning outcomes or platform engagement.
	H ₁	Higher instructor proficiency significantly predicts better student engagement and perceived learning value.
Role Perception & Motivation	RQ7	To what extent does students' perception of the LMS as a passive file repository suppress motivation?
	H ₀	Whether students perceive the LMS as a file repository or active learning environment has no significant effect on engagement.
	H ₁	Students who perceive the LMS merely as a content repository will demonstrate significantly lower engagement, motivation, and interaction.

Table 4.2: Summary of Hypotheses

Chapter 5

Result Analysis

5.1 Qualitative Findings

This part is an analytical report of the semi-structured responses of the participants who are undergraduate students, graduate students and faculty members of various Bangladeshi Universities. and other institutions of higher learning. The interviews examined the lived experiences of students with Learning Management Systems (LMS) with regard to usability and instructional support. Using the thematic analysis, seven interconnected themes were formed illustrating essential usability issues with usage patterns that affect LMS use and efficiency in a developing-world higher education setting. The results combine the views of the participants with verbatim quotes in order to explain how the LMS design, practices in the platform, and institutional circumstances affect the learning experience of students and their daily academic activities.

5.1.1 Diversity In Platform And Fragmentation

Students in various institutions reported to use more than one digital platform simultaneously to accomplish various academic tasks and thus it has become a disjointed experience in utilizing technology learning instead of a cohesive, integrated LMS space. Students switch institutional platforms, communication tools and external learning services instead of depending on a single centralized system. One participant explained, *"Since I am a BRAC University student, I use BUX, Google Classroom, Slack, and sometimes Discord. But most of the time, I use Coursera."*(J2). This was the case in interviews, including *"For academic purposes, I mostly use BUX (our university system). I also use Discord, Slack, Google Classroom, and sometimes Udemy and Coursera."*(A3) and *"Slack, BUX, Google Classroom, and Discord."*(A5). Students from other universities reported comparable arrangements,

including *"We generally use two platforms: Microsoft Teams and AIUB's own portal."*(A4) and *"Well, I am a student of North South University, and the LMS that we use is Canvas Student."*(T1). Collectively these descriptions reveal that students do not often find themselves in a position to perceive their academic platforms as a single system; rather learning activities and information is spread over a multiplicity of disconnected tools. This disjointed setting is influenced by the institutional necessities as well as instructor predilections that make the students on a case-to-case

basis diversify their working processes. Participants stated that there are those platforms which are compulsory and others are informally chosen by the faculty. One student noted, *"Some are voluntary and some are institutional. Faculty members choose between Discord or Slack, and BUX is required for course enrollment and accessing materials."*(A3). Another similarly stated, *"Both. Sometimes it is used because it is required, and sometimes we use it voluntarily."*(A1). This can lead to platforms becoming narrow in function as opposed to full learning environments. As an example, one of the participants wrote, *"Many Coursera courses provide Discord server links, so I use Discord for discussion and communication."*(J2), while also clarifying, *"For Discord, I would say it is mainly for communication, not for teaching and learning content directly."*(J2). Other students echoed the same separation, such as *"I mostly use Google Classroom and sometimes Discord for group discussion with my friends, but mostly for academic purposes, I use Google Classroom."*(T3) and *"I mainly use Google Classroom, Discord and Slack for communication with the faculties."*(T6). This indicates that students in their minds divide platforms into their tasks (materials, submissions, discussion) and not as an integrated system of learning. Students also were aware of the negative aspects of maintaining

multiple platforms simultaneously and found it obvious to be desirable to consolidate. According to one student's observation, *"At first, it was confusing. However, I think using external tools is necessary because one platform cannot handle everything."*(A5). Others proposed a more structured solution, stating, *"I think it would be better to use one core LMS and integrate other platforms with it."*(A5) and *"Having one core LMS that integrates other platforms would be the most helpful."*(A5). These reactions indicate that fragmentation may be viewed as a workaround and not an ideal arrangement, and that learners appreciate systems that are capable of minimizing platform switching and at the same time enhance communication, content delivery, and academic coordination under the same roof.

5.1.2 User Interface And Usability Concerns

Students defined usability as what they experience after using it everyday, how easy it is to use, by how clear and laid out it is as well as how easy it is to use to find what is required. Platforms that put emphasis on communication were usually termed as easy to use and empathetic. One participant stated, *"I think Discord is very user-friendly."*(J1), while another explained, *"Coursera's interface is very user-friendly. It is usually clear where I am, where I should go, and what I am doing. Everything feels self-explanatory."*(J2). Furthermore, we also perceived usability because of the visual comfort and aesthetic design. As an instance, one of the students remarked, *"Since I use Discord the most, I would describe its interface as dark-themed and visually appealing, with user-friendly fonts and layout."*(A1). Meanwhile, some

respondents shared unpleasant experiences with institutional LMS platforms, the language they used tends to be confused and irritated. One participant said, *"No, it is not clear. Sometimes it is confusing."*(S1), and described the reasons as *"Slow loading speed, too many steps, and it is visually unappealing."*(S1). These conflicting reports indicate that the choice of elements in the platform design, both visual and functional, determine whether the students find LMS as an easy and user-friendly

platform or a stressful and ineffective one to use. One of the same worries of all in-

terviews was how learning materials are organized and how to effectively search and retrieve such information. Students reported problems with locating course materials in cases when it was poorly organized or when the platforms had to be scrolled at all times. One participant explained, *"Teachers usually provide study materials in a single place, but the materials are often not well organized, so it becomes difficult to find things."*(J1). Others also gave such reports, such as *"The main issue is needing to scroll through long lists of content."*(J2) and *"It lacks a search function, and teachers' materials are disorganized."*(S1). By making certain that a platform was easy to operate in general, the disorganization complicated the use of the platform, even when students regarded it as easy. As is expressed in the statement, *"Yes, I often face problems finding course material because the files are not well organized and I have to scroll a lot to find new content."*(T7). Contrary to this, websites with

sections that were visibly delineated were said to be simpler to handle. One participant stated, *"It has separate sections for files, assignments, and chats for different teams, which makes it organized."*(A4), and reinforced the same point by adding, *"The separation of files, exams, and assignments helps a lot."*(A4). These analogies imply that usability is closely related to information architecture: students can get away with simple interfaces, but they are not able to work with interfaces where the content is not always organized and readily available. Student feature overload

and interface density were also characterized as working impediments particularly when the platforms are full of tools that do not seem to have a relationship with academic requirements. One participant observed, *"Yes. I think the 'Nitro' premium features are unnecessary. For example, users can add extra GIFs or custom profile images, but these features are not needed for academic use."*(A1). Other people explained how excess amounts of channels and notifications may make platforms seem busy and difficult to navigate, such as *"Sometimes there are too many channels in Discord. There are many section-based channels that I have to mute, otherwise I receive too many notifications. I think the interface should be cleaner, with fewer sections."*(A3) further emphasizing *"Yes. Some Microsoft features feel overwhelming because there are too many of them."*(A4). Although students tend to

get accustomed to it with time, the common issue is that with more lucid designs, more topical sets of features, and more efficient organization of materials would help save the amount of needless effort and streamline the LMS usage process in daily study.

5.1.3 Technical Challenges and Infrastructure Boundaries

Technical performance and infrastructure restrictions were constantly mentioned by students as daily any inconvenience that could disrupt learning processes, particularly when it comes to time-sensitive activities like submission, access to materials, or online classes. Problems of file manipulation were especially widespread. *"File size limitations are a major issue."*(J1), and further explained, *"When we try to send multiple project files together, we often face file size restrictions."*(J1). Another

student also pointed out, *"Limited file size."*(S1), clarifying that the difficulty is especially visible during uploads, *"File size limits, especially for PDF uploads."*(S1). Such limitations are a source of frustration when conducting coursework, which relies on the exchange of huge documents or the sharing of various files, and it appears that easy academic tasks are not so easy. Other than file handling, students often com-

plained about slow system performance, lag and system instability. One participant summarized their experience as, *"The main problems are slow loading, server issues, login problems, and sometimes assignment upload errors."*(T4). Others described similar slowdowns in their daily use, including *"It becomes slow sometimes."*(T3), and *"But there are some issues whenever I'm posting something or any assignment on Google Classroom, given tasks, if that takes a bit of time, that means it becomes slow."*(T3). Students using Canvas reported delays that affected awareness of course

activity, such as *"The canvas is a bit slow when there is notification."*(T1), and *"The notification just doesn't get updated as fast as I think it would."*(T1). Access and continuity were also interrupted by authentication issues. One participant explained, *"Our canvas is not actually logged in all the time. It logs us out after every one or two days."*(T1), while another recalled, *"One issue that I faced once was my canvas student ID was having some issues because I did not log in for a long time. It was giving an error."*(T1). Such problems delay the process, lack of consistency in sessions and failure to log in generate confusion and lower reliability particularly when students rely on the platform to meet deadlines, make uploads or announcements. They also included the use of resources and platform weight especially when machines might have less RAM or processing power. One participant recommended, *"First, it should use less RAM and be lighter to run."*(A1), and added, *"Reducing RAM usage would have the biggest impact, because it would make the platform smoother and easier to use during study."*(A1). Similar concerns appeared among Microsoft Teams users, including *"It should be less congested with Microsoft features, it feels too heavy."*(A4), and, *"They should improve RAM usage and change the notification sound."*(A4). In addition to the technical performance, affordability

also became a viable infrastructural issue. One participant noted, *"Some LMS platforms that previously offered free access are becoming paid."*(J2), and spoke out the impression made by experience: *"Earlier I could use Coursera's financial aid and earn certificates without paying. But now I need a paid monthly subscription."*(J2). They emphasized the local economic context clearly: *"For Bangladeshi students, even 10or15 can be a lot."*(J2). On the whole, these replies demonstrate that reli-

ability, efficiency, and affordability are tightly interconnected with the capacity of students to utilize LMS tools in a resource-starved environment with a smooth and consistent flow.

5.1.4 Learning Curve and Training Deficiencies

The vast majority of the students noted an initial learning curve during the first time using the LMS platforms, but this difficulty level was dependent on the platform and their previous technical experience. Initial problems were usually related to naviga-

tion and the search of materials. One participant stated, *"In the beginning, it was difficult to find study materials."*(J1), while another pointed to access barriers from the start, *"The login process was difficult because it had too many steps."*(S1). It also vastly differs on various platforms in terms of initial learning. A statement by one of the participants captivates that, *"It depends on the platform. Discord was very complicated at first, but once you become familiar with it, it becomes quite easy."*(A5). The other respondent talked about beginning to use LMS in the middle of the pandemic, noting that the lack of knowledge made the use of the tool challenging in the first place, *"During COVID, before my admission test, I first started using an LMS at Udvash. At first, it was difficult because it was my first experience with such a platform. It took some time to get used to the interface."*(A3). Conversely, other

platforms were said to be simple to embrace at a given moment, particularly when the students were already familiar with the digital world, like. *"Google Classroom was very easy to learn because it is user-friendly."*(A2), and *"It was easy because I have been using computers for more than 10 years."*(A1). Such diverse descriptions are indicative of the fact that the learning curves are influenced by both platform design and digital confidence of the students. Despite this confusion in the early stages, students said they improved with time, and as a result of repetition instead of formal training. Some of the participants reported gradual acquaintance, including, *"Over time, after Discord updates, it became easier to use."*(J1), *"It improved over time. I can use many features comfortably now."*(A1), more stated that *"Yes, it improved over time."*(A4). One more participant summed up a more general normalization effect: *"Gradually, I learned it, and now most LMS platforms feel similar."*(A3). Nonetheless, students stressed many times that such an improvement tends to oc-

cur without institutional onboarding. There were numerous accounts of the lack of formal training, including: *"I learned it by myself."*(J1), *"I did not receive formal training. I watched YouTube videos and learned on my own."*(J2), and *"None. There was no support."*(A4). The most frequent ones were informal assistance of colleagues and self-investigation, which were mentioned in such words as. *"Mostly I learned by using it by myself, and sometimes by asking friends or seniors."*(T4), and *"The support I received was mainly from my fellow mates who were using it."*(T2). What this implies is that the students tend to emerge as effective users regardless of the system, but not due to systematic support. The absence of on boarding and training relates to the bigger problem of long-lasting engagement and regular use of LMS as well. Other participants gave accounts on changes in motivation and usage patterns over time, such as *"At the beginning, I used Coursera daily and completed coursework regularly. Later, my activity reduced."*(J2). Other accounts gave an overview of an overall learning culture of trial and error, stating, *"I think it's the same for everyone. We don't actually get any help, we just go around the app and see what there is in it."*(T1). Such dependence on self-learning is a cognitive

load that can put at a disadvantage students who start with a lower level of digital literacy. Although the majority of the students eventually become accustomed to the platforms that they consume, the interviews indicate that the implementation of LMS is usually based on the assumption that individuals will learn on their own, and training and orientation is the unsolved issue in the practice of institutions.

5.1.5 Notification and Communication Effectiveness

Students repeatedly stressed that notifications and communication tools are not the additional opportunity, but a core of how they can be aware of coursework, deadlines, and updates provided by the instructor. Clarity and timeliness of notifications on LMS platforms give students more control over their learning and allow them to act more quickly in response to the tasks. As an example, a participant reported that reminders and alerts facilitate the process of engagement directly, *"The notification system is very helpful."*(A1). A different person related the deadline view to practical gains, stating, *"There is a given due time so we can submit it anytime from our home."*(T3). In such situations, the notification systems serve as usability aids by eliminating uncertainty and aiding students of the course in the management of coursework without the need to continuously search the platform. However, communication and notification effectiveness becomes weak when systems are either unreliable or overwhelming. Late notification delivery was also reported by the students and makes them less trustful and compelled to control the platforms by hand. Highlighting from one of the student's statements, *"The notification just doesn't get updated as fast as I think it would."*(T1). The reverse of the issue was also reported

by other people: excessive notifications, too much noise, and messages overload. As one participant noted, *"For Discord, I feel like there are too many notifications. I muted the notifications and many messages get lost as all the students message and use Discord for queries."*(T6). Such overload drives students into coping mechanisms such as muting that unwillingly introduces the risk of missing valuable academic material. Students also described how important files and links can disappear in fast-moving communication spaces: *"For Discord, messages move first and files and links get lost if they are not pinned."*(T6). These reactions demonstrate that the design of the notification should be aimed at achieving the speed, simplicity, and priorities; otherwise, the communication platforms will be chaotic, and the LMS will not run as a stable academic setting. Because of these problems, students often rely on external tools for communication even when LMS tools exist. One participant stated clearly, *"We actually prefer messenger groups or WhatsApp groups, these types of things as students."*(T1). However, students also pointed at the fact that

effective collaboration is raised by powerful communication tools. One participant explained, *"Using Discord, I can connect with them, chat, call, and even arrange meetings."*(J2). Structured communication is cited as another cause of clarity by another: *"In Discord, teachers can create separate channels, which makes communication very clear."*(A3). When communication works consistently, it even influences motivation, as one student stated, *"As there is continuous communication with the teachers and professors, I would say that it definitely motivates students."*(T2), and another summarized the overall impact: *"It does motivate and helps students learn, getting materials, reminding the deadlines and it helps us to interact with everybody at ease."*(T7). In general, the results indicate that the efficiency of notification and communication is directly related to the differences between students who use the LMS as an effective learning platform and those who use it as a tool that they have to avoid through the use of external applications.

5.1.6 Instructor Proficiency and Its Impact

Students defined the proficiency of instructors as a firm matter that determines whether an LMS would develop into an orderly learning atmosphere or a disorganized place that would make students work even harder. Multiple participants noted that the experience of instructors becomes better over time, particularly in cases when platforms are new. One student explained, *"At the beginning, they did not have enough knowledge because it was new to them."*(J1), but later added, *"Over time, they became familiar with it through repeated use."*(J1) and *"Now, I think they have adequate knowledge."*(J1). This trend indicates that the proficiency of instructors is dynamic and is built upon exposure and practice. Students also compared institutional systems and platforms where the instructors are seen to be better prepared. As an illustration, one of the participants mentioned, *"Coursera selects instructors carefully. They appoint capable instructors."*(J2), which is the reflection of the quality of instruction and system experience that may influence students' confidence in the platform. Meanwhile, several students reported that there is in-

consistent proficiency among members of the faculty, as some teachers operate the LMS well, whereas others do not or do little. One participant stated, *"Not everyone, but most of them do not."*(S1), and another associated teacher practice with general involvement through observing, *"The LMS should be more user-friendly, and it should improve student engagement."*(S1). Some respondents mentioned that instructor inactivity is a significant obstacle even with a platform that works. One student explained, *"Some instructors are not active. They do not share consultation timings or provide guidance."*(A5) and emphasized the consequence, *"If the instructor is not active, the LMS does not work effectively."*(A5). They further clarified that this problem is not necessarily technical, *"I do not think this is a fault of the LMS; it depends on the instructor."*(A5). This underscores the importance of usability results in the design of the system as well as the manner in which the instructors put the system into practice. In cases where the instructors were experienced and

regular, the students noted obvious advantages, particularly in the feedback, efficiency, and support with the learning. One participant said, *"Yes, I think they are very proficient."*(A1), and explained how that proficiency helps, *"I cannot give an example where it harmed my learning. But for improvement, instructors often share useful tools and code snippets through Discord messages, which helps me learn and improve my skills."*(A1). Similarly, another student stated, *"The grading system is very efficient. Instructors can grade assignments directly through the LMS, so students do not need to visit them in person. This improves efficiency."*(A2). There

was also course-by-course variation, according to which students reported learning differently based on the chosen platform and habits of the faculty. According to one student they stated that, *"In some cases, no. I have seen new faculty members who cannot use Discord or Slack properly."*(A3), and described inconsistency across courses: *"In one course, I had no communication problems, but in another course, the instructor used WhatsApp instead of email or LMS tools."*(A3). The implication in the summary made by the participant was as follows, *"I think some faculty members need training."*(A3). On the whole, teacher competence is one of

the determinants of LMS effectiveness, and the experience of students demonstrates that regular institutional guidelines and the willingness of the faculty can mitigate platform fragmentation and enhance its use.

5.1.7 Role Perception and Motivation

The perception of the purpose of an LMS by students had a strong influence on the degree of reliance on it and its role in motivating and engaging the student. According to many participants, LMS platforms were mostly functional but stored resources and facilitated access to them, rather than being active learning environments. As an illustration, one interviewee explained the importance of connection and centralized access by saying, *"It connects teachers and students on a single platform, and they can share materials easily. That is very helpful for learning."*(J1). Another student explained motivation in similarly practical terms: *"Yes, I think it motivates students because they can access all learning materials there."*(J1). Mo-

tivation in these descriptions is connected not to enjoying things but to a decrease in friction students are inspired when systems facilitate learning content. Nevertheless, a number of students defined the use of LMS as conditional or limited, which is often due to the lack of meaningful learning to drive day-to-day learning other than file storage. One participant said, *"I use it when I need it; otherwise, I do not. I am not fully dependent on it."*(S1). Others described LMS platforms mostly as repositories, such as *"Mainly as a content repository."*(A2) and *"Mainly for accessing reading materials and checking class schedules."*(A5). A student described the role very narrowly, *"Well, I do think the LMS gives us a place where all of our lectures are stored. It is a place where each student often finds their lectures. That's about it, actually."*(T1). Similarly, another participant said, *"Mostly, our slides and PDFs are uploaded there. So that's how we just get the offline materials from our classes. So we just follow that. That's the main way I think."*(T3). The responses indicate that in the situation where the LMS is perceived as storage only, it is a passive medium, instead of a learning environment that is functional in providing engagement. Meanwhile, other students found LMS platforms as a

repository and as a communicative space, particularly in the case of appropriate use of communication and feedback tools. One of the participants explained in his own words, *"I see it as both a content repository and an active learning environment."*(A1), while another said, *"I think it serves as both."*(A4). Another aspect

that was mentioned by students was the change in the role of communication and feedback, which makes the system more active. As an example, one participant described a value of feedback, *"Discord is helpful for receiving feedback. Teachers can point out mistakes, and students can improve for the next exam."*(J1). A different student explained Coursera as a more holistic learning role, *"For Coursera, it is a strong tool for both learning and communication."*(J2) and elaborated, *"Content is available in one place, students can access previous materials, and they can ask questions to instructors in specific sections."*(J2). Students frequently reported a

bifurcated system of roles in the institutional context, with the various platforms

addressing varying needs, *"Google Classroom helps manage the course as there are sections such as tasks, files and feedback. Whereas Discord helps for communication such as discussion, group support and quick questions and answers."*(T6). Under these circumstances, the LMS ecosystem is capable of facilitating motivation, yet it also reinforces fragmentation since no particular system is able to execute all of the functions effectively. Generally speaking, it appears that role perception and

motivation are heightened where platforms are readily available, offer feedback, reminders, and articulate communication to make the LMS simply feel like a living learning community, as opposed to a dead storage.

5.2 Statistical Analysis

5.2.1 Qualitative Results Data Analysis

Raw qualitative data from the interviews were coded initially using codes that identified patterns and conceptual labels which matched the participants responses. All of these codes were organized systematically and presented in a structured code-book that would be used as a common reference for all coders. Each coded entry was independently viewed by four raters. Differences were recorded in the rating columns which explained clearly in the data. To check for consistency of coding, Krippendorffs Alpha was used as a statistical measure of agreement. the $K - value = 0.92$ which represents a high degree of consensus between raters. The verified codes were then analyzed conceptually and grouped into clusters. From this seven broad themes were inductively derived from these clusters of codes, which were based on the supporting codes,.

A thematic analysis carried out on all 34 interview participants (including students enrolled in undergraduate programs from various universities in Bangladesh, faculty members, and alumni) identified a pattern that was consistent and widely spread across participants with regard to LMS-related usability issues. Most amount participants were hassled by the following issues. In case of User Interface and Usability 98.2% participants were in line with this context. notification and communication effectiveness were reported by 97.4% of the participants. And role perception and motivation as shown in Figure X. All these are emerging themes that highlight their core importance in creating the overall LMS experience, irrespective of institutional affiliation, platform preference, and user role in the Bangladeshi higher education context. The second most common theme identified 91.2% was Instructor Proficiency and Its Impact on Learning, which highlights general concerns about the digital competency of teachers and the downstream effects on engagement and quality of course delivery. 88.2% of participants discussed learning curve and training deficiencies, reflecting a lack of institutionalized support for structured on boarding, as the majority of users, both students and faculty, find themselves on unfamiliar platforms and take it upon themselves to learn how to navigate them and operate within them and 77.6% of participants indicated that most users relied on self-directed or peer-assisted learning to navigate uncharted platforms. 79.4% of the participants reported technical challenges and Infrastructure Boundaries, including problems with slow system performance, size of files, session timeouts and unreli-

able notification delivery, and found these to be meaningful usability challenges in resource-limited situations. Lastly, platform diversity and Fragmentation was the least common theme identified (73.5%), but it was still an issue raised by nearly three-quarters of all respondents, suggesting that the concurrent use of multiple, non-integrated platforms is still causing redundancy, inconsistencies, and cognitive overload in the learning landscape.

The findings taken together validate that the usability problems faced by LMSs in Bangladesh are not isolated to technical issues but rather are integrated into the interface design, communication infrastructure, institutional support structures and pedagogical role expectations.

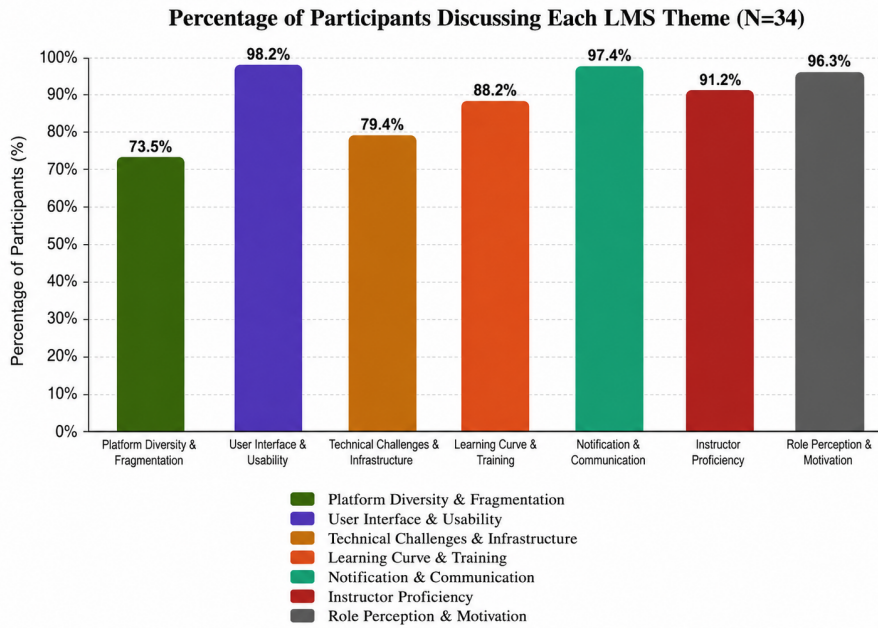


Figure 5.1: Prevalence of LMS usability themes among participants

5.2.2 Quantitative Results Data Analysis

To get our outputs of the surveys and to verify our themes with generalized data we needed to exemplified theoretically in H2 and thus demonstrated empirically: Q1 chi-square test on platform count significant (Q1): $p < 0.0001$; versus Q2 t-test of disruption intensity : not significant alone (Q2): ($t = 0.118$, $p = 0.453$). However, this hypothesis is strengthened because the distributional evidence from Q1 (that students do not sit equally across the various categories of platform use – 3 or more simultaneously) is evidence of this feature. Results from the t-test do confirm, however, that the magnitude of disruption perceived is found very close to the neutral point [$M = 3.01$] and this is described as a fine grained finding that will be further expanded upon in H2 Interpretation.causes navigation complexity and issues. Having a greater effect on student engagement than an intuitive interface does and lack of that result of less engagement from the students. For interface issue checkboxes, with the use of a chi-square test, it was revealed $\chi^2(4) = 41.053$, $p < 0.0001$ along with "More steps to complete tasks" which was mostly reported ($n = 105$), followed by "Hard to find materials" ($n=86$). The avoidance of navigating was

not uniformly distributed across frequency categories $\chi^2(4) = 217.955, p < 0.0001$. H_0 is rejected. These are consistent with the qualitative finding that cluttered paths and non-searchability hinders LMS engagement.

In case of, H1 (UI and Navigation) assessed whether poorly designed interfaces and which causes navigation complexity and issues. Having a greater effect on student engagement than an intuitive interface does and lack of that result of less engagement from the students. For interface issue check boxes, with the use of a chi-square test, it was revealed $\chi^2(4) = 41.053, p < 0.0001$ along with "More steps to complete tasks" which was mostly reported ($n = 105$), followed by "Hard to find materials" ($n=86$). The avoidance of navigating was not uniformly distributed across frequency categories ($\chi^2(4) = 217.955, p < 0.0001$). H_0 is rejected. These are consistent with the qualitative finding that cluttered paths and non search ability hinders LMS engagement.

H2 (Platform Fragmentation) empirically examined whether multiple digital platforms are linked to increased academic friction when using these three or more at the same time. There was a highly significant $\chi^2(3) = 91.261, p < 0.0001$ chi-square for platform-count that confirms the non-uniform and multi-platform use bias of the respondents. The disruption t-test was perfectly non-unique ($t(221) = 0.118, p = 0.453$) with a sample mean of $M=3.01$ which falls near the neutral midpoint where both dimensions meet, but chi-square rejects H_0 anyway! What we have found in our t-test is that although students are objectively divided on platforms, there is some mixed experience of disruption in their subjective experience, probably when they feel the same way when they are exposed to a population of Bangladeshi students who are normalizing their platform switching behavior.

H3 (Technical Infrastructure): Technical problems are common and can prevent users from using LMS. The most often reported disadvantages were file size limitations ($n = 93$) and a slow loading process ($n = 92$). A chi-square on problem distribution showed that $\chi^2(5) = 32.973, p < 0.0001$. There was an interconnection between Yes/No interference question and The severity t-test, which would yield $t(171) = 1.765, p = 0.040, M = 3.12$ for 172 respondents (77.5%) who responded "Yes" to the interference question, suggesting a statistically significant difference above the neutral point of the scale. H_0 is rejected for all three sub-tests.

H4 (Training and Institutional Support) evaluated the impact of lack of formal training on LMS mastery. For the learning-method chi-square, $\chi^2(4) = 62.730, p < 0.0001$. The distribution is heavily biased towards the informal learning: 78 respondents reported using YouTube or online tutorial as a source, 65 relied on trial and error whilst only 14 stated formal institutional training. H_0 is rejected. The long answer open text question (LWW), Q3, is analyzed thematically, not statistically, and this is fed into the qualitative discussion. This is a sign of the lack of institutional on boarding of LMS in our sample of universities, and more likely the lack of such on boarding than its presence.

H5 (Notification and communication) wondered if there were any problems with notifications preventing students from staying on LMS-based communication tools. The

chi-square for the communication platform is $\chi^2(15) = 1318.685$, $p < .0001$; with email ($n=153$), and Discord ($n=113$) being the two most used channels. A chi-square analysis revealed a notification effect $\chi^2(4) = 87.730$, $p < 0.0001$. H_0 is rejected. The world outside of the traditional LMS has also gained importance in the academic communication process; Discord and WhatsApp/Messenger have been clearly enlarged in contrast with the embedding into the formal LMS process.

H6 (Instructor Proficiency): Applied teacher use of the LMS as a predictor of student use of the LMS and student value of learning. We use one-sample analysis with the proficiency rating t-test, which yields: $t(221) = 3.056$, $p = 0.001$, $M = 3.19$ this means that ratings are significantly above the neutral midpoint and that students perceive their instructors as moderately but meaningfully proficient at most since they feel their instructors are more proficient at fewer than half of the items of interest in many competencies/tasks measured here. The chi-square comparing the influence of instructor activity on usage resulted in $\chi^2(2)=10.784$, $p=0.005$, indicating that the majority of students attested that instructor LMS activity does have an influence on student own LMS usage. H_0 is rejected. This finding agrees with the qualitative assertion of moderating effect of the instructor behavior in the adoption of LMS.

H7 tested the hypothesis that the perception towards the LMS as content repository is correlated with less engagement and motivation in students. The primary use chi-square had a value of $\chi^2(5) = 21.587$, $p = 0.001$, with assignment submission and file download being the two most common uses. The t-test of motivation yielded was significant, $T(221) = -2.399$, $p = 0.009$ with $M = 2.83$ which is significantly lower than the neutral midpoint confirming that LMS motivation is suppressed. The chi-square for the interaction terms yielded $\chi^2(2) = 65.216$, $p < 0.0001$. Any one of the three tests result in rejecting H_0 . It is particularly notable that the below-midpoint motivation mean of 2.83 provides direct quantitative support to the qualitative result that students view the LMS as a resource, not a classroom, as this phenomenon was found in this school.

Hypothesis	Tests Used	Min p-value	α	Decision
H1 - UI & Navigation	Chi-sq tests	$p < 0.0001$	0.05	Reject H_0
H2 - Platform Fragmentation	Chi-sq & t-test	$p < 0.0001$	0.05	Reject H_0
H3 - Technical Infrastructure	Chi-sq tests & t-test	$p < 0.0001$	0.05	Reject H_0
H4 - Training & Support	Chi-sq	$p < 0.0001$	0.05	Reject H_0
H5 - Notification & Communication	Chi-sq tests	$p < 0.0001$	0.05	Reject H_0
H6 - Instructor Proficiency	t-test & Chi-sq	$p = 0.0013$	0.05	Reject H_0
H7 - Role Perception	Chi-sq tests & t-test	$p < 0.0001$	0.05	Reject H_0

Table 5.1: Hypothesis Results

5.3 Analysis of Design Solutions By Prototype

This design specification will convert our qualitative interview evidence for concrete, inspectable UI patterns for South/South-East Asian contexts. While it is reusable, it goes beyond typical usability metrics and tackles the ground problems inherent to our regions, such as platform fragmentation, low-bandwidth accessibility, and institutional training gaps, bridging user needs with practical, appropriate solutions in different cultures. Each feature integrated in our prototype supports our area of contribution, usability problems, and the design principles of Norman.

To address the Platform Fragmentation we have input some features in our offered prototype. Firstly, we have integrated a Unified Student Dashboard, which will help students to get easier access to their course materials, notification etc within just 2 clicks while satisfying a unified surface. Secondly, the integration of the Top Navigation bar remains the same throughout the prototype. This establishes a design foundation for an integrated surface and keeps the role symmetry. Thirdly, we have Course hub with public and private contents. Through these features the students from other universities will also be able to use the contents which are public and the contents which are set as private will be locked and displayed to them as blurred. Through this we have made all course tasks discoverable. This helps with chat-style channels to opt out of redundancy of “Discord detours” and it maintains mapping and constraints from Norman’s Design principles. Lastly, we Global search to search across materials, messages, announcements etc. The results show the sources and the date through keyboard-first navigation. Therefore, with a single entry point it replaces cross platform hunting. All these features serve for the Norman’s Design principle of conceptual modeling, mapping, visibility and feedback. Through all these features we have addressed the Platform Fragmentation part within our prototype.

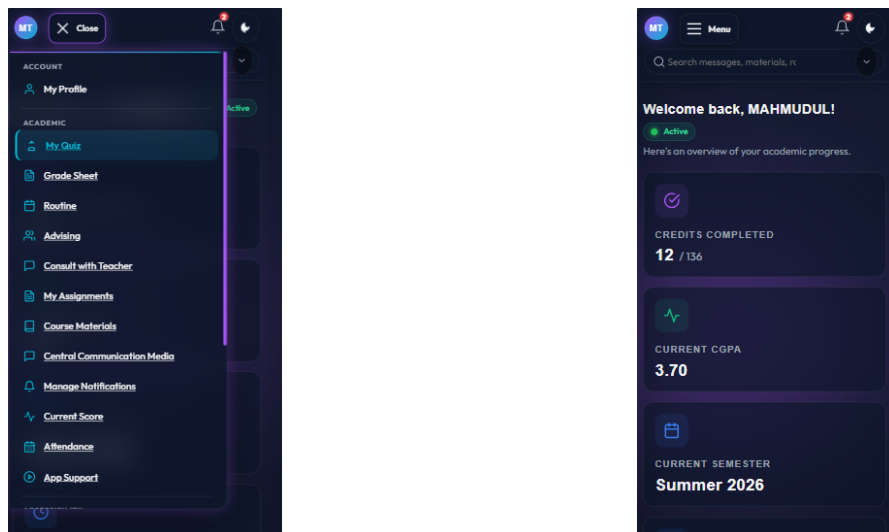


Figure 5.2: Unified Student Dashboard and Landing Page

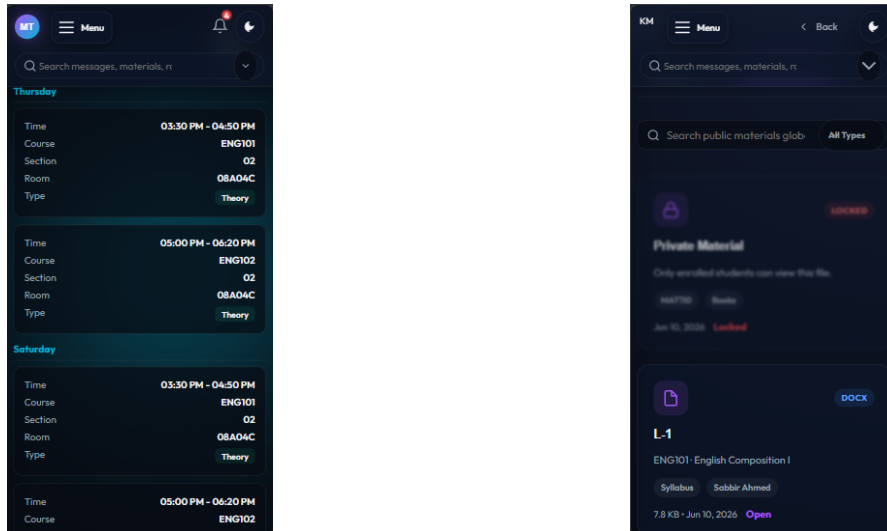


Figure 5.3: Landing Page with Routine and Access to Public and Private Materials

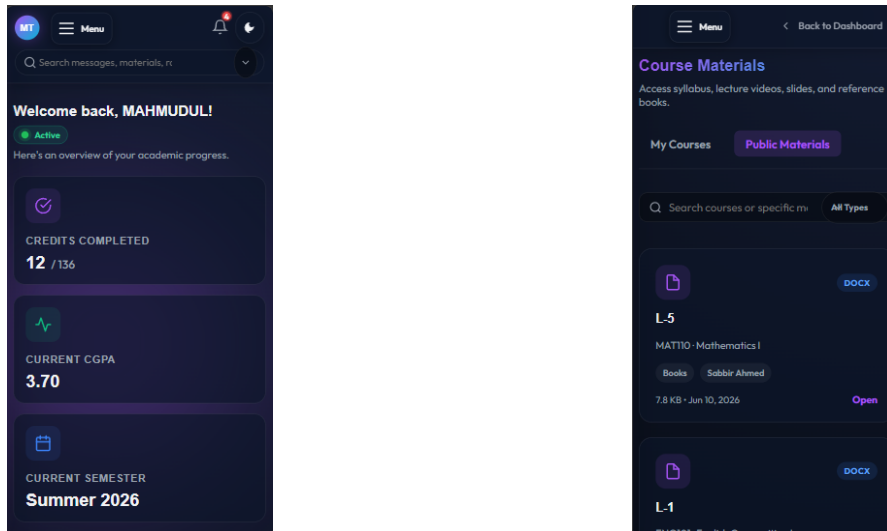


Figure 5.4: Global Search Bar with Nav Bar and Course Materials Repository

Then comes UI & Usability concerns and to address that we have integrated multiple features. Firstly, we have the Materials tab with scoped search. This feature helps to find anything within 5s, including filter chips (slides, reading materials etc.). Through this feature we are fulfilling evidence-backed organization patterns for low-friction retrieval. Secondly, we have integrated Material cards, which displays icons, size, date, and an inline preview. This provides lightweight file handling clarity and also reduces unnecessary downloads. As preview opens inline thus items get first priority to render and recently viewed strip stay persists. Lastly, we have Light/dark theme and density toggle, ensuring keyboard focus states and high-contrast tokens. This helps with density persistence while focus stays on rings evident and redefines accessibility for low-resource contexts. All these features maintain affordance constraints, mapping, visibility and feedback from the Norman's Design principles. This is how we have addressed the UI & Usability concerns part within our prototype.

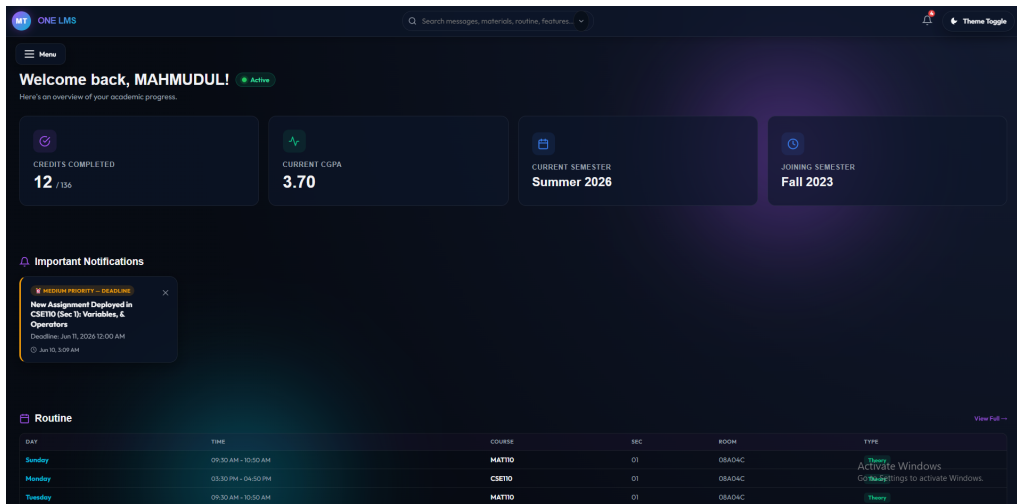


Figure 5.5: Working Prototype across devices

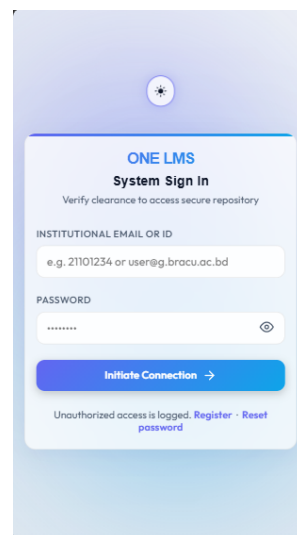
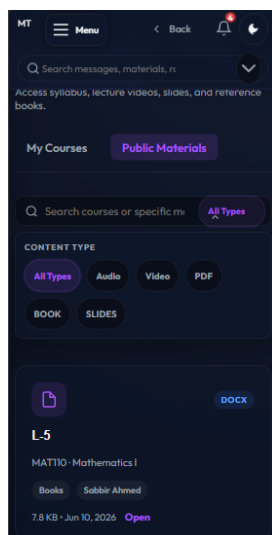


Figure 5.6: Materials Tab and Light Theme

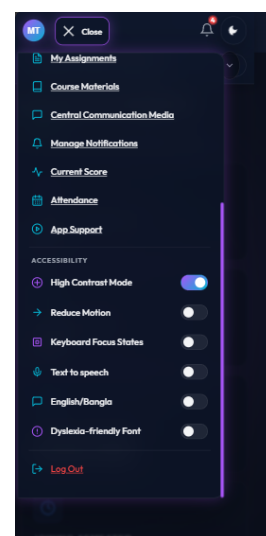
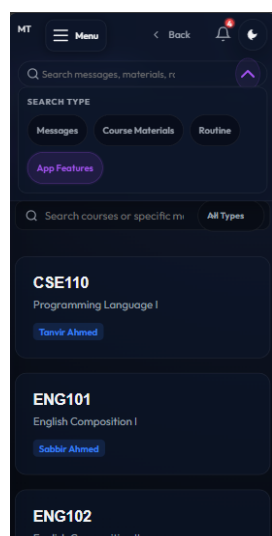


Figure 5.7: Global Search Bar in Mobile and Accessibility Features

For Notifications Overload and Communication Effectiveness, we have integrated multiple features to address these. For the Notifications Overload part, firstly we have the Notifications center where per-item gets priority badges and the mentions, deadlines, announcements etc. are prioritized as tabs. Through this feature, badges can be immediately updated with “Mark all read” and it takes only one single click to show the deadlines. Thus, to manage overload, this feature delivers smart, filterable notifications which will be in the favor of the users. Secondly, we have integrated Contextual preferences which allows per-type notification preferences. This toggles a channel or type and reflects data state with badges. This feature replaces mute-all workarounds with context-aware control. Lastly, we have Deadline reminders to establish practical, evidence-backed reminder timing. This simulates alerts 2-3 days ahead of due dates and also allows widget color-codes deadlines, such as red is for less than 24 hours, yellow is 2-3 days, and green more than 3 days. All these check out affordance, constraints, visibility and feedback from Norman’s design principle. For the Communication Effectiveness part, we integrated the Course Discussion feature. This feature has multiple channels such as announcement, theory section, Lab section, Queries along with channel based search, @mentions, message threads etc. All these offer an integrated Discord but also allows experience to curb fragmentation. This interesting feature falls in the affordance, visibility and feedback category from the Norman’s Design principles. This is how we have addressed the Notifications Overload and Communication Effectiveness within our prototype.

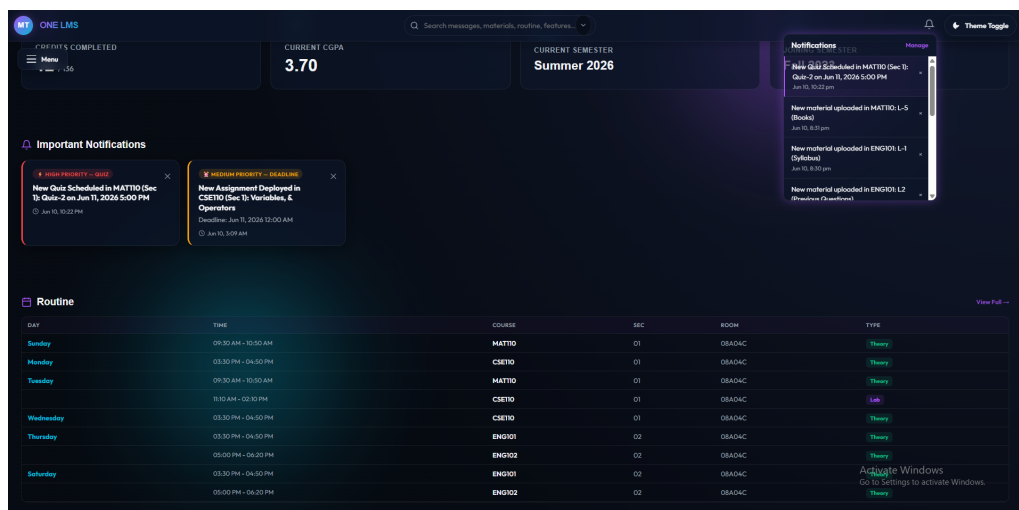


Figure 5.8: Priority Notifications And General Notification

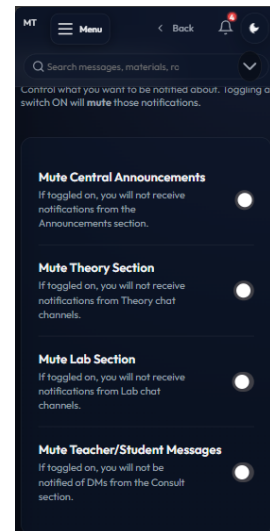
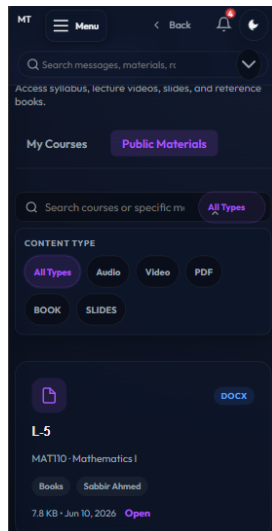


Figure 5.9: Bell Icon For Notifications and Notification Management System

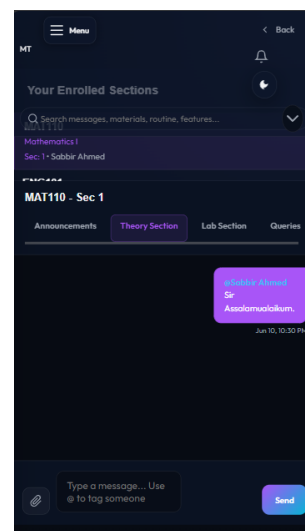
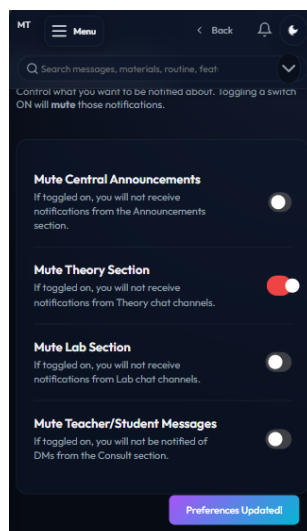


Figure 5.10: Notification Controls and Central Communication Media

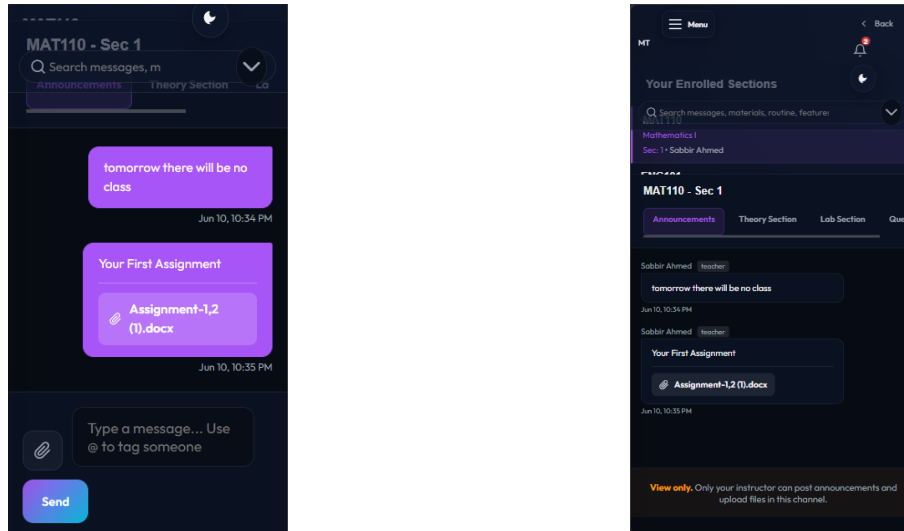


Figure 5.11: Central Communication Media Teacher and Student View

Then we have addressed Training Deficiencies in several features. First we have introduced 3 video tutorials located in the app support, which contains the tutorials on how to use this entire app. These videos show search, deadlines, channels, and filters on single click, which are saved locally for future guidance. Thus, this introduces institutional training as a variable. Then we have integrated a Communication media for contacting with the app developers which will help the users to directly communicate with its developers and get instant solutions and to learn how this app is used. It is linked from the menu bar and it stays state persisted in per sessions open or close operation. Through this self-serve guidance is encouraged and reliance on peer-only support is lowered. These features maintain conceptual modelling, mapping and visibility from the Norman's Design principles. This is how we have addressed the Training Deficiencies part within our prototype.

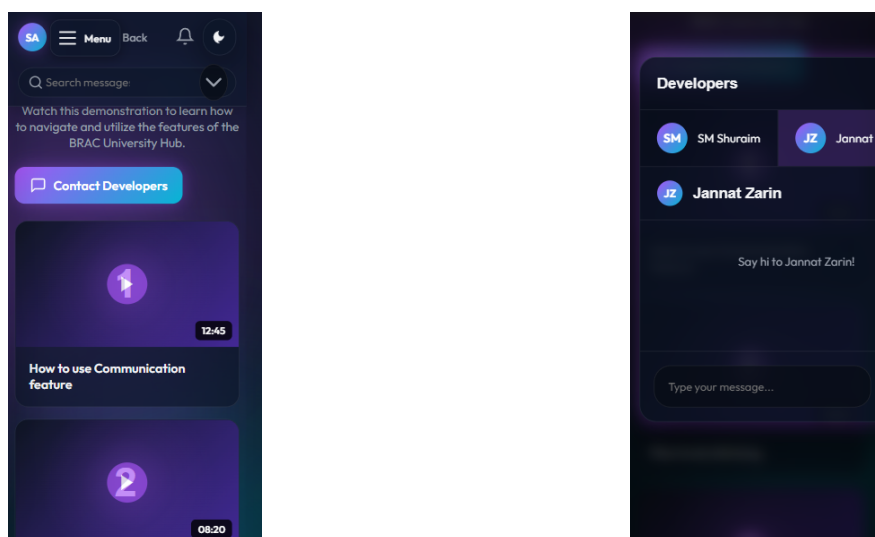


Figure 5.12: Video Tutorials of App Controls and App Support Through Developers

We have also worked on the Instructor Proficiency and Feedback Effectiveness in our prototype. The app further supports role based login and view support. If a person logs in with a teacher account they will be displayed and will be able to use the features of a faculty and if anyone logs in as student they will be able to view and use the features of a student account and finally if anyone randomly creates account other than the institutional account they will be considered as guest and they will be able to access all the public resources available in our app and get help in their academics. The faculty view exposes a dedicated dashboard for publishing quizzes, assignments, grades, view pending or submitted quizzes and assignments of students section wise of the course they are teaching, course management controls, and a grade viewer for assignments submitted to the faculty. Faculty members post announcements, upload materials, and create assignments via straightforwardly mapped action routes, each bearing a confirming success banner that reflects immediately on the student side. This feature helps the instructors by lowering the action cost. This interesting feature falls in the affordance mapping and feedback category from the Norman's Design principles. Thus Instructor Proficiency and Feedback Effectiveness have been addressed in our prototype.

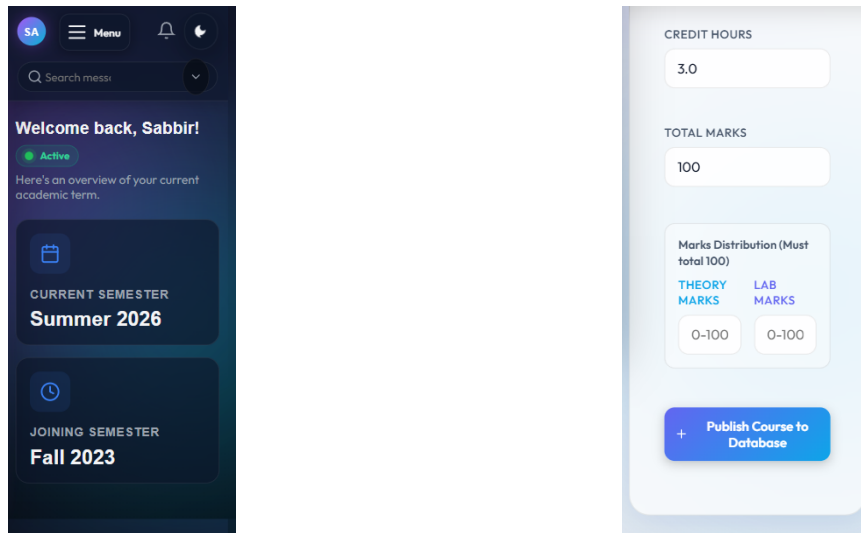


Figure 5.13: Teacher Homepage and Teacher Creating Courses

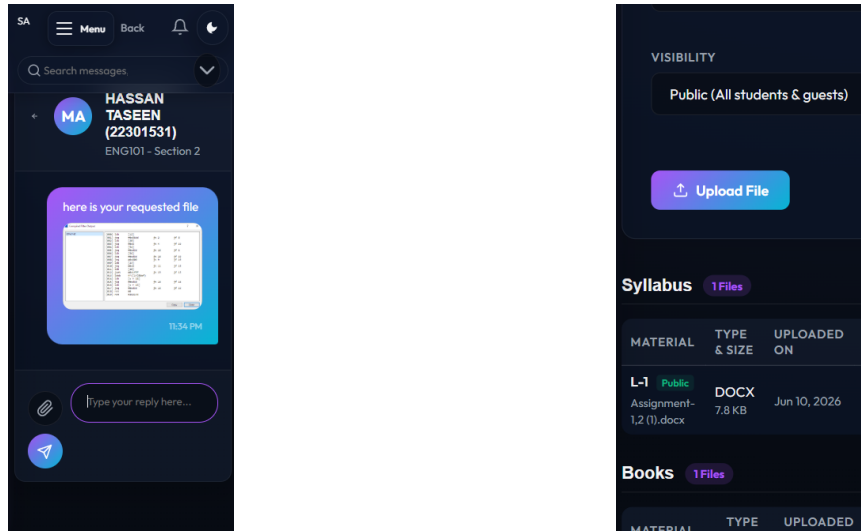


Figure 5.14: Teacher Student Communication and Teacher Course Content Control

Then we have also addressed the Technical Constraints and Performance Perceptions with Persistent session indicator. It allows no surprise logout messaging and also user friendly re-auth cues. Due to this feature, the header shows “Session: Active” shows a clear banner. This feature addresses the trust fall caused by sudden session drop. While this feature maintains the feedback category of Norman’s Design principles, this is how we have addressed the Technical Constraints and Performance Perceptions in our prototype.

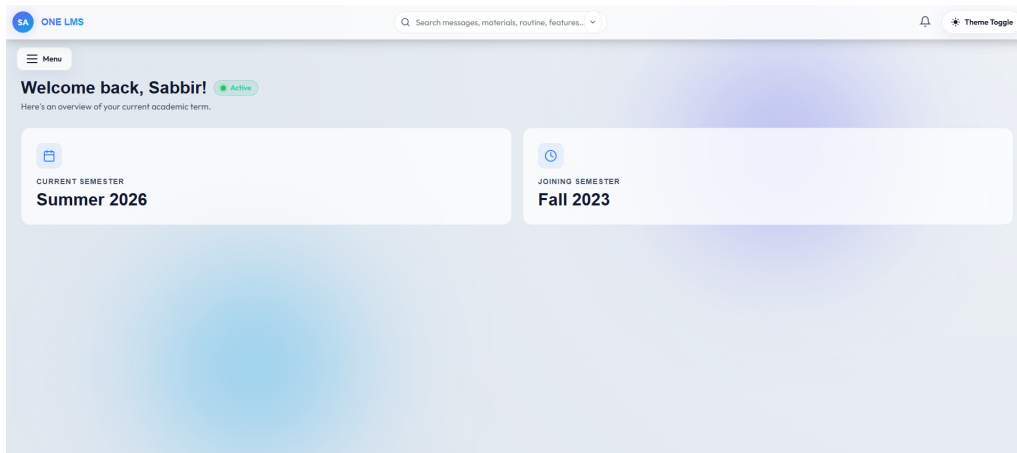


Figure 5.15: Active Session Feature

Finally, to address Motivation and Role Perception we have integrated Menu button in the Dashboard. Due to this feature the prototype has Advising, Assignment Submission and deployment, Quiz submission and deployment, grade-sheet preview and submission, Communication and direct consultation button placed on the Menu directly. This saves the time of platform hopping while nurturing active behaviors. This interesting feature falls in the affordance and visibility category from the Norman’s Design principles. This is how we have addressed the Motivation and Role Perception part within our prototype.

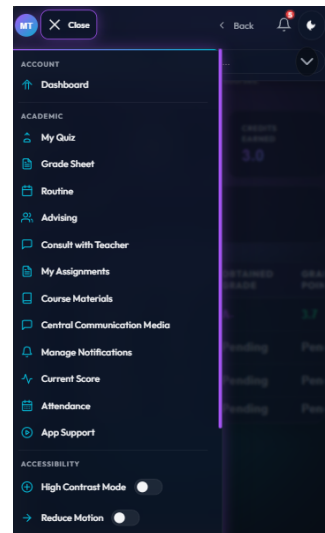
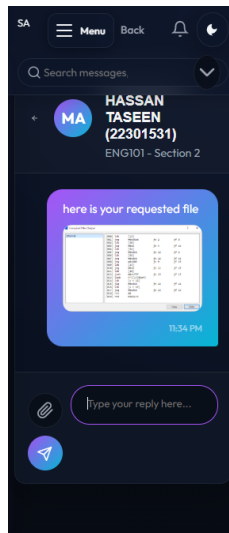


Figure 5.16: Communications and Student Drawer

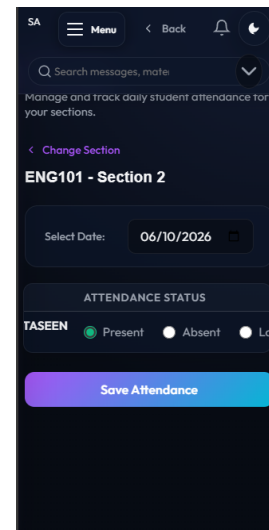
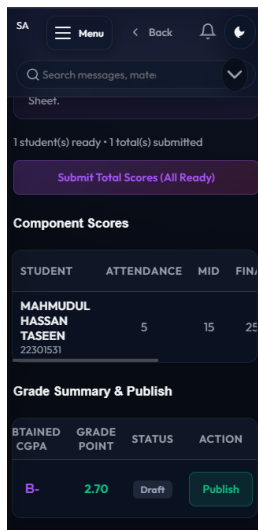


Figure 5.17: Grade Submission and Attendance Control

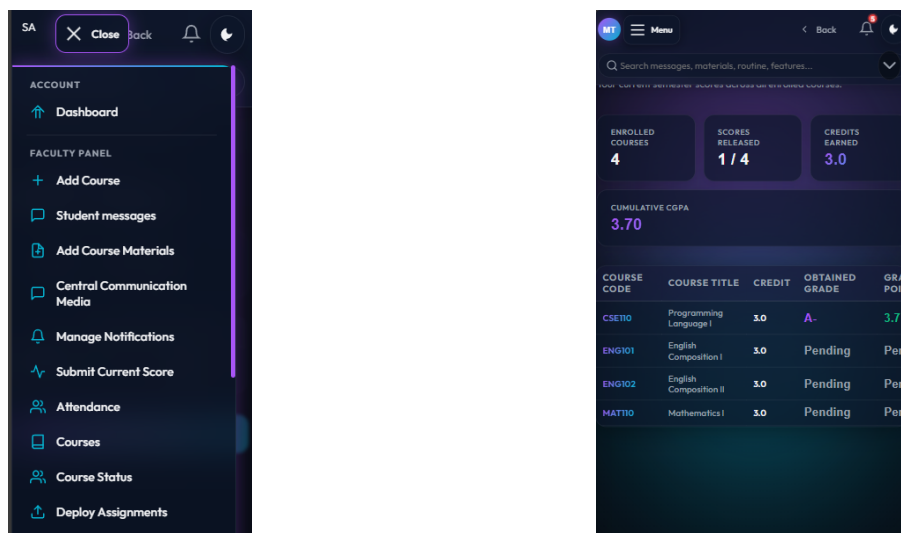


Figure 5.18: Teacher Specific Menu and Course Content Control

To sum up, this LMS prototype overcome technological and usability issues in higher education among South/Southeast Asian. This prototype helps to mitigate all the concerns found during studies through a user friendly and easier approach without being overstimulated. Alongside this suggested prototype followed Norman’s design Principles and Universal Design Principles to ensure digital equity in low-resource environments. Thus this research is an approach to effectively bridge spatial requirements with practical UI solutions.

5.4 Comparisons and Relationships

In Bangladesh based universities operate their courses on a variety of platforms: Canvas at NSU and a number of private universities, Microsoft Teams at AIUB and others, Google Classroom nearly everywhere due to being free, Discord and Slack informally used by course groups, Moodle at most public and many private universities, Zoom as the live classes, and buX, the own platform of BRAC University, which is based on Open edX. The issues listed here are based on all the public user feedback, vendor support forums and published studies which are listed in Sources. This is a brief summary; the table below disaggregates the same information by platform.

The first group of complaints is regarding weight and speed on standard hardware. Microsoft has admitted itself that Teams can consume approximately a gigabyte of memory when idle and runs slowly on modest laptops [58] [39]. Moodle has been repeatedly claimed to be slow and clunky, particularly on mobile and low-quality connections [47] [45]. and a public bug report against buX demonstrates an embedded lecture video pushing a laptop CPU to work for operations. It in a few seconds [48] . These are not trivial inconveniences to students on tight budget machines and lopses in internet, which are the standard in Bangladesh.

A second group is regarding structure and findability. In Canvas, the assignment,

its due date, the grade and the work submitted are in different locations, requiring one to constantly switch tabs, and each instructor sets up a course in a unique way [35] [33] . Moodle and Google Classroom render materials truly difficult to find, and there is no powerful search within courses [30] [47] [31] . Discord hides files in a rapid chat and does not grade files at all [46] , whereas Slack hides files more than 90 days old and deletes files more than a year old [42] [43] [22] . Google Classroom as well considers work as missing when a final step is not taken, which deduces marks off students [26] [12]. The teams overwhelms students with emails about assignments that cannot be turned off [40] and Moodle notifications are usually delayed or never received [47].

A third group is regarding access and fit. The live classes provided by Zoom on top of any LMS a university may already have disconnects students with a poor internet connection completely and does not provide any form of course structure itself [9] [37] [34] . Discord, a platform designed to socialise and play games, distracts students and has the potential to contain unrelated or even adult material [41] [32] [46] . buX is a purpose-built application that operates on an outdated MOOC-style platform with discussion disengaged from the content, was publicly glitchy upon release, and continues to be a distinct system alongside Zoom and Email [6] [7] [10].

We have answered these in our prototype. The dashboard (with a colour-coded upcoming-deadlines panel and a quick-actions sidebar) and course pages (with tabs), a materials library, assignments, a calendar and discussion channels, all are located in a single course structure, so nothing is spread across tools. Locating is managed with a global search and course scoped search with type filter chips and preview. The assignment flow maintains a countdown based notifications with file upload and a persistent post-submission banner. so there is never any doubt about whether it submitted. A notifications center including priority tabs and per-type and per-course preferences allows students to manage alerts without deadline reminders and a persistent session indicator ensures they are not caught off guard by a logout. The interface remains light, featuring a consistent top navigation menu, a light/dark theme, a compact density setting and an English/Bangla language switcher tour. Each fix corresponds to a feature in the prototype, with the sources of the problems mentioned at the bottom.

Platform	Key design problems reported	How our LMS addresses it	Sources
Canvas (NSU and other private universities)	No visible indicator when a session is still active. ; assignment details and deadlines are inconsistently presented. Each instructor structures courses differently, making materials hard to locate; pages load slowly.	Persistent session indicator. Single assignment detail , upload, confirmation banner, feedback system. Uniform materials page with search, filter chips, view toggle, preview; global search; fixed top navigation; App support.	[44][64][35][29][33]

Platform	Key design problems reported	How our LMS addresses it	Sources
Microsoft Teams	Assignment notification emails cannot be disabled without also disabling calendar alerts. Office-first interface cluttered with chats, channels, and tools irrelevant to academic use.	Notifications centre with priority tabs and per-type/per-course preferences; deadline alerts retained. Academic-only interface with simple top navigation, role-based views.	[58][39][40]
Google Classroom	Submitted work can appear as missing; students lose marks with no clear indication. Poor organisation; no cross-course search. Minimal gradebook and limited feedback options.	Persistent post-submission banner make status unambiguous. Course-scoped and separate global search, and filter chips Discussion boards, calendar, and announcements.	[26][12][30]
Discord	Documents and links buried in fast-moving chat with no organisation. No built-in assignments, grading, or read receipts. Social/gaming context encourages distraction and server-switching.	Dedicated searchable materials library independent of chat. Complete assignment workflow (create, submit, grade, feedback) with grade-pending view. Academic-only platform with role-based views.	[41][32][46]
Slack	Free plan hides messages and files older than 90 days; files over one year are deleted. No course structure, assignments, gradebook, or grade tracking. Requires managing another platform alongside email and the institution's LMS.	All content stored permanently in an organised materials library. Tabbed course pages, dashboard due-date view, calendar, and complete assignment and feedback workflow. Single institutional platform replacing fragmented tools.	[42][22]
Moodle (most public and many private universities)	Outdated, cluttered interface with steep learning curve and excessive clicks. Materials hard to locate; each course laid out differently. No clear view of completed work; slow or absent notifications.	Uniform top navigation, clean dashboard, theme switch and FAQ. Uniform materials page with search, filter chips, view toggle; global search. Colour-coded deadline alerts, operable routine, notification center with priority tabs; English/Bengali toggle.	[47][31][36][45]
Zoom (default live-class too)	Meeting-only tool with no course pages, assignments, gradebook, or progress tracking. Extended live sessions cause fatigue; losing a login loses access to session content.	Core academic work requires no live video. Single integrated structure: course tabs, materials, assignments, routine, announcements, and discussions; persistent session.	[9][37][18][34]

Platform	Key design problems reported	How our LMS addresses it	Sources
buX	Outdated browser-specific base; features fail in modern browsers. Discussions separate from coursework; platform runs in parallel with Zoom and email rather than replacing them.	Modern browser-agnostic interface with light/dark themes and consistent layout. In-course discussion channels, materials, assignments, feedback system, calendar based routine, and announcements unified in one place.	[48][6][7][10]

Table 5.2: Comparison of platforms.

5.5 Discussion

Our study analysed the majority of LMS usability issues in Bangladeshi higher education system’s Learning management systems. By this we understood how usability deficits learner’s engagement and methods of teaching. During our interviews through major universities in Bangladesh we discovered seven interconnected themes that were collectively in conflict with many of the usability issues of the LMS systems in Bangladesh which were challenged within a developing-world context. These findings of ours shows that LMS usability related issues are not majorly shortcomings but are failures that are interconnectivity systematic failures. For example, platform fragmentation among systems and gaps in training programs are notably visible issues. Other than that, notification misalignment and competent instructor judgment are also major flaws. All these factors together are shaping the interaction between students and LMS platforms. Which directly affects the students’ engagement levels. So these findings point straight at our study’s core objectives where we examine LMS usability within resource-constrained higher education environments.

From our analysis we got that students effectively and regularly navigate between three to seven platforms as their daily tasks require. This represents a significant usability barrier that is particularly straight forward in developing-economy contexts. Though Annamalai et al. [11] and Al-Fraihat et al. [54] focused on the use and adoption of LMS in institutional mono-platform settings, this paper shows a significantly different picture where platform proliferation is not voluntary but a part of the structure. This fragmentation occurs primarily from institutional management mishappenings rather than pedagogical intent or user preference. This kind of system pushes students to acquire and manage multiple credentials and learn disparate interfaces. Also with these comes a barrage of micro management tasks like monitoring competing notification systems and locating course materials distributed across different platforms without any centralized coordination. As one participant explained, *“Some are voluntary and some are institutional. Faculty members choose between Discord or Slack, and BUX is required for course enrollment”*(A3). Also this kind of uncoordinated adoption pushes students to assume the role of platform managers rather than learners. This load of fragmentation is further heightened in environments that have bandwidth restrictions and loading delays, which add to mental load of switching among systems. While some students got accustomed

to this like, *“At first, it was confusing. However, I think using external tools is necessary because one platform cannot handle everything”*(A5) they simultaneously also proposed systemic solutions, *“I think it would be better to use one core LMS and integrate other platforms with it”*(A5). These findings expand upon the work of Behnam Faghieh et al.’s [1] framework by showing that even well designed platforms become functionally complex when embedded within fragmented ecosystems. Students tend to organize platforms based on their functional capabilities, *“For Discord, I would say it is mainly for communication, not for teaching and learning content directly”*(J2); *“I mainly use Google Classroom and sometimes Discord for group discussion”*(T3) preventing holistic engagement with LMS. Our research objective examines these deficits and addresses LMS adoption issues. Student’s conditional use patterns like, *“I use it when I need it; otherwise, I do not”*(S1) indicates suppressed adoption agendas driven by systemic division rather than individual resistance.

Furthermore, students show recurring frustration over absent search functions and disorganized materials which also correlates to our prior findings emphasizing the importance of interface design in LMS usability [1] [54][61]. Nevertheless, this study brings to light the fact that there is even more to this problem in that course materials are often uploaded in the order in which they were created and cannot be found by searching except by the expert users. This is not just the limitation of platform design but also the lack of institutional and pedagogical spaces of information architecture practices. Participants repeatedly stated like, *“Teachers usually provide study materials in a single place, but the materials are often not well organized, so it becomes difficult to find things”*(J1); *“The main issue is needing to scroll through long lists of content”* (J2); *“It lacks a search function, and teachers’ materials are disorganized”*(S1). Despite long term uses of LMS platforms with 62.5% reporting 3 to 4 years of use and 31.3% reporting over five years, it still persists that students cannot navigate over the LMS environment which depicts further on systemic design flaws and management failures rather than user unfamiliarity. While discord were praised for their appearance over interface, *“dark-themed and visually appealing”* with *“user-friendly fonts and layout”*(A1). Students also noted down functional constraints and the irritation over message loss. Conversely, more reliable institutional platforms were criticized as *“visually unappealing”* and cumbersome(S1). From this disconnect we can analyze that solely aesthetic quality does not support the fact that it’s less or more usable particularly when information architecture and instructional practices are weak. Which furthermore emphasizes the need of a structured organization and this can be validated by, *“separate sections for files, assignments, and chats”*(A4) which were deemed more usable and interaction friendly, but their organization were more dependent on instructor behavior rather than platform affordances. This is in line with the information gaps that were identified by Dzandu and Tang’s [2] as a result of unclear structure and lack of contextual information. Students stated that the materials frequently were uploaded without any explanatory frame, which restricted the assimilation of knowledge between sessions. Nevertheless, overload on features also added over the previous problems of usability challenges. Participants criticized unnecessary premium features *“the ‘Nitro’ premium features are unnecessary”*(A1) and excessive channel creation *“Sometimes there are too many channels in Discord”*(A3). Which suggests there should be more specific task oriented design and instructional discipline which are more important

than expanding more feature sets.

Infrastructure constraints became one of the identifying usability issues that differentiate this study and LMS studies in the developed economies. The participants informed of slow loading and failure of the system. Also inability to log in and file upload errors like, *“The main problems are slow loading, server issues, login problems, and sometimes assignment upload errors”*(T4). Such failures are not just inconvenient, as they undercut the interaction in LMS as a platform to do high-stakes academic activities. File handling capabilities were also encumbered, *“File size limitations are a major issue”*(J1); *“Limited file size...especially for PDF uploads”*(S1). These limitations break the workflow of coursework and show that the decision making of platform design in the context of resource rich environments imposes limitations in the developing economies. To further hurt the cause there was authentication instability over prolonged sessions which further hurt the engagement process. Students described frequent logouts and access errors, *“It logs us out after every one or two days”*(T1). Which as a result hampered the use and discouraged reliance over LMS platforms. These issues directly address the concerning adoption rates which indicates in what ways infrastructural unreliability is suppressing prolonged and sustained engagement. We noticed that device oriented limitations are a critical but under examined factor. Participants emphasized the need for platforms that, *“use less RAM and be lighter to run”*(A1), echoing similar concerns about Microsoft Teams, *“They should improve RAM usage and change the notification sound.”*(A4). This extends upon previous research work on connectivity inequality [56] by highlighting hardware limitations as a form of digital hindrance. Other than that, affordability also came into question. One participant observed, *“Some LMS platforms that previously offered free access are becoming paid...For Bangladeshi students, even \$10 or \$15 can be a lot”*(J2). These financial barriers further complicate LMS adoption in limited resource environments.

A very important systemic failure that was identified in all four universities was the absence of institutional training. Even after using LMS platforms for years after years, participants reported about self learning, *“I learned it by myself”*(J1), *“None. There was no support”*(A4). So the findings of Sabir and Ali et al.[24], Karim et al. [56], and Saarmean et al. [16], are confirmed and expanded by this. This demonstrates that, even after the pandemic period, the training gap prevails. It was also noticed that students mostly depended on peer support, *“by asking friends or seniors”*(T4). So it gave inconsistent and fragmented learning. The importance put on social norms by Revyathi and Tselios’s [4] was reflected by this informal transfer of knowledge above. Importance was put on social norms while disclosing its limitations. Peer based learning keeps up the incomplete system understanding. Then again, the participants highlighted the wasteful trial and error learning, *“we just go around the app and see what there is in it.”*(T1) The lack of a well organized structure of institutional support mechanisms was reflected by these findings. This also directly mentions the objective of the research by testing the development of professionalism and digital literacy in the adoption of LMS.

Communication and notification systems became the focus of student engagement. But it had a tendency of not working most of the time. On the one hand, the timely

notifications enhanced the control “*The notification system is very helpful*”(A1) on the other hand, the timescales harmed the trust, “*The notification just doesn’t get updated as fast as I think it would*”(T1). Too many notifications were overwhelming and distracting, “*I muted the notifications and many messages get lost*”(T6). The absence of information in high-speed communication media “*messages move fast and files and links get lost*”(T6) encouraged students to use external services like WhatsApp and Messenger(T1). This is an indicator of institutional LMS communication failure. In cases where the communication tools worked, they helped in boosting the motivation and interest, “*it definitely motivates students*”(T2). The results make a direct association between communication usability and the engagement of the learners and the effectiveness of the teaching.

The competence of the instructors was found to be a decisive modulation factor that influences the effectiveness of the LMS. Although a few instructors acquired competence with time (J1), majority were seen as poorly competent, most of them do not (S1). Lecturer inactivity “*If the instructor is not active, the LMS does not work effectively*”(A5) lowered platform value to a considerable extent. In other situations, students controlled LMS spaces because of limitations by the instructor, “*students often manage it*”(A3), which is an inversion of pedagogical roles. On the other hand, instructors with competence facilitated learning by making good use of communication and grading instruments. These results support the validity of the fact that the design quality of LMS cannot be complete without the corresponding instructor capacity.

The majority of students have viewed LMS platforms as content repositories, “*That’s about it, actually*”(T1); “*Mainly as a content repository*”(A2). Practices by the faculty supported this view as they used the LMS as a means to share files only. This inactive use is contrary to the purpose of the research of creating interactive and collaborative learning conditions. Nevertheless, other students perceived LMS as multifunctional tools when they actively developed the features of communication and feedback. The motivation was also not similar, and some students have claimed that, “*the LMS itself does not motivate me to learn*”(S1). These results indicate that interaction depends on pedagogical behaviors and not only on platform capabilities.

In our study it validates the prior studies over LMS usability and extends upon it by revealing distinctive challenges in developing economy contexts. Koca and Özcan [57] conducted mixed-methods research with teachers at a single university, finding moderate SUS scores (63.85) indicating considerable usability issues. Their studies pointed toward problems like poor infrastructure and lack of support with minimal management. However, Koca and Ozcan [57] noted the “*lack of student’s voices,*” a limitation explicitly addressed in our study through a focused analysis of student experiences across different universities. Similarly, in the research of Simon et al. [62] it examined Moodle through mixed-methods research with 41 teachers and 41 students, finding that most users saw Moodle as little more than a file sharing system based on their usage. By this study we can also see that the scope across multiple platforms in bangladeshi contexts and which also correlate with the observation that any kind of formal training was largely absent. That’s why our research demonstrates how the absence of training still continues even after years post

pandemic era. As a result, suggesting systemic institutional failure rather than temporary crisis conditions. Berényi et al. [52] assessed the usability of Moodle among officials of Hungarian universities with the System Usability Scale (SUS) with a high result of mean SUS scores (85.04). However, the students were not satisfied with notifications and collaboration tools despite a high level of satisfaction with quizzes and login operations. This is in line with Lu, Schmidt, and Shin [59] who criticized the most popular reliance on general usability measures such as SUS on the grounds that these tools do not take into account pedagogical and sociocultural elements. Lu, Schmidt, and Shin et al. [59] created the Comprehensive Assessment of Usability Scale of Learning Technologies (CAUSLT) that embraces the technological, pedagogical and sociocultural aspects. Their criticism is further confirmed by our study's findings, the Bangladeshi students are happy with the simple features of the platform, but at the same time they complain about the systemic failures in the training infrastructure and the overall institutional assistance. We also noticed that Ngulube and Ncube [60] conducted a systematic review of specific studies leveraging analytics to enhance LMS experiences. They found that while dashboards and machine learning tools were regularly used, despite that the adaptive capabilities of students remained low. Concisely, they concluded that paying too much attention to the technical side of matters may not identify the underlying issues of educational activity and that proves analytics alone cannot facilitate meaningful pedagogical change [60]. This aligns perfectly with our statement that technology-focused solutions are insufficient without institutional capacity, governance, and pedagogical alignment.

Adding to that, Global comparisons also highlight the situational results. A study by Almusharraf [28] investigated the perceptions of Saudi students about Blackboard, the results showed that the reliability and importance were rated higher than the usability and accessibility and the level of experience affected the priorities. Which builds upon that observation in Bangladeshi settings, where students have currently made infrastructural reliability and institutional backing their highest priority of concerns and that the interactivity of the platform is inconsequential when the access is not reliable. Also, Dzandu and Tang [2] highlighted that there are missing contextual infrastructure because of poorly managed materials in LMS systems which clashed with knowledge acquisition, creating semantic and syntactic gaps. By correlating with their framework Bangladeshi students not only face poor interface design but also failures over structure which should be based upon profound information. Which discloses the fact that course materials are uploaded and managed without sufficient context or organization.

Moreover, We can see in works of Secaira et al. [61] that to have a significant LMS evaluation, quality measurement in a multi-dimensional and context sensitive approach is critical. This research proves their claim normative usability measurements cannot be adequate in times when infrastructural integrity is at such a state where alarm is raised. On top of that, access and engagement in resource limited environments limits institutional and pedagogical factors.

Lastly, There are a few characteristics that the Bangladeshi higher education situation can be differentiated with other developing and developed economies. Segmen-

tation of the platform is more of an institutional governance failure than a choice of the user or a conscious pedagogical choice. Exclusion is worsened by device-level resource limitations (e.g., RAM limits that do not support the use of LMS on older devices), which supplements previous studies on internet connectivity disparity [56]. The lack of formal institutional training continues to exist in several universities, many years after the pandemic, as a structural inability in IT governance and instructional design experience, and not a short-run emergency deficiency [16]. All this makes the LMS ecosystem a very delicate system with infrastructure, competence of instructors, and organizational support highly moderating factors related to adoption, engagement, and teaching effectiveness. The results sufficiently cover all the mentioned research aims showing the impact of the lack of usability on the adoption rates, engagement of learners, and instructional practices. LMS effectiveness is determined by platform fragmentation, unreliability on infrastructure, lack of training, and competence of the instructor. Although the qualitative scope makes generalization impossible, the results are very informative in terms of context. Further studies need to combine quantitative and qualitative methodologies and should involve faculty and administration. Importantly, institutional, pedagogical, and governance changes instead of technology-centered ones are needed to implement LMS in emerging economies.

Chapter 6

Conclusion

In Bangladeshi higher education, LMS is grossly underutilized, though large institutions have invested heavily in it. Here it serves as a file repository as opposed to a learning ecosystem. It is now as clear as a crystal that inferior design makes everything worse. When technology is hard to use in education, it not only baffles users but also intensifies all the impediments to the existing ones. In Bangladesh, there are many students who don't have stable internet access, or don't have access to proper devices and sometimes do not have proper digital literacy. For them navigation becomes more tough if the platform is also poorly designed. This does not serve the purpose of democracy of education as the educational technologies backslides the core claim. Moreover, sometimes instead of improvement the hardship increases and this adds more adversity to the deprived ones. These invisible usability blockades are critical to creating unbiased and essential learning exposure for all students, only when they are understood and addressed.

Recent studies revealed several fundamental shortcomings which have to be in the limelight. It is a matter of great concern that the majority of research operates within a Western-centric paradigm. This completely overlooks the pedagogical, sociocultural, and infrastructural realities that characterize education in developing countries. System Usability Scale (SUS) has an important role in the field, though it does not emphasize the effectiveness of education and the cultural context. Moreover, in the research landscape there can be seen domination of cross-sectional studies rather than longitudinal validation. Therefore, we are failing to understand the evolving process of usability issues. It is also noticed that student feedback is getting consistently marginalized and a persistent gulf between identifying problems and developing solutions. Therefore, it can be said that the issues are being documented but there is rare actionable interference. These disparity are rising in Bangladesh as there is a lack of mixed-methods study examining LMS usability considering lack of digital literacy, insufficient training infrastructure, and unreliable technological resources.

This research paper specifically fills these imperative gaps with context-related mixed-method appraisal. Having both quantitative usability measures and rich qualitative insights from both students and faculties, this ensures understanding of all feedback. After the documentation of the issues, our contribution also lies in developing feasible evidence-based design recommendations within Bangladesh's re-

source constraints. These meaningful recommendations are bound to bring tangible change in the system. The intermittent disconnection between the identification of a problem and the practical solutions are also bridged through an integration of pedagogical effectiveness, sociocultural nuances and infrastructural realities altogether. This study is designed in such a way that it will give a practical framework for Bangladeshi institutions to transform their LMS, which will be an inclusive learning space for students of all walks. On a larger scale, we hope such localized practice can be adopted by other developing nations, which are experiencing equally difficult situations in their digital learning experiences through context-based research.

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6.1 Appendix

Prototype

The interactive prototype is available at: [URL for Prototype](#)

Codebook

The codebook (XLSX) is available at: [URL for Code book](#)